

3D Volume Image Analysis for Biologists

Fiji / Napari for classical Image Analysis methods

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Content

- Multidimensional images
- Fiji:
  - Stack manipulation and exercises
  - 3D image reconstruction and visualization (volume & surface rendering)
  - An application study: 3D image analysis
  - Tracking
- Napari:
  - Basics and manipulation
  - Stack manipulation and exercises

Images can have many dimensions

The number of dimensions is the number of things you need to know to find each pixel

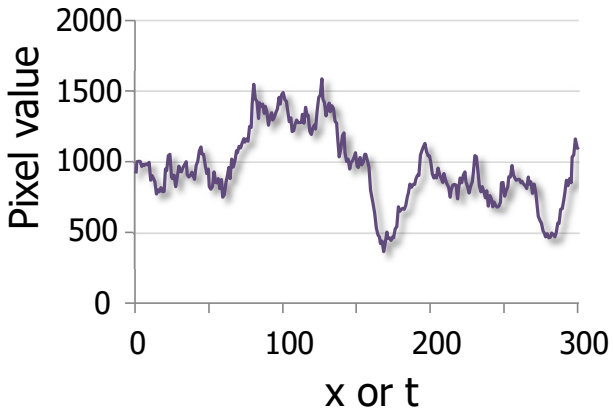
17

To find a pixel, you need:

0 dimensions
(just a single pixel)

Slide courtesy: P. Bankhead

Images can have many dimensions



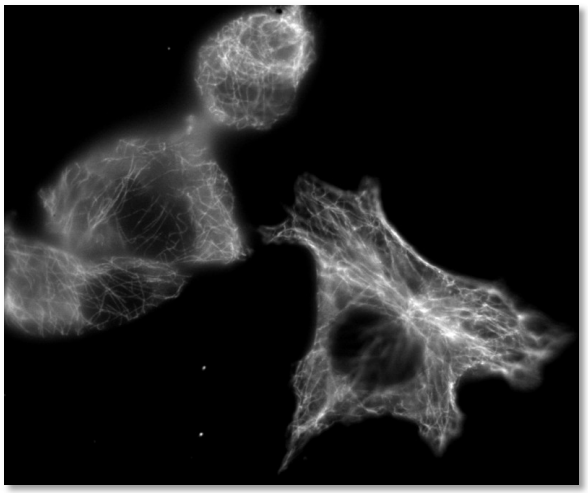
1 dimension
(can be plotted as
a graph)

The number of dimensions is the number of things you need to know to find each pixel

To find a pixel, you need: x coordinate (x)

Slide courtesy: P. Bankhead

Images can have many dimensions



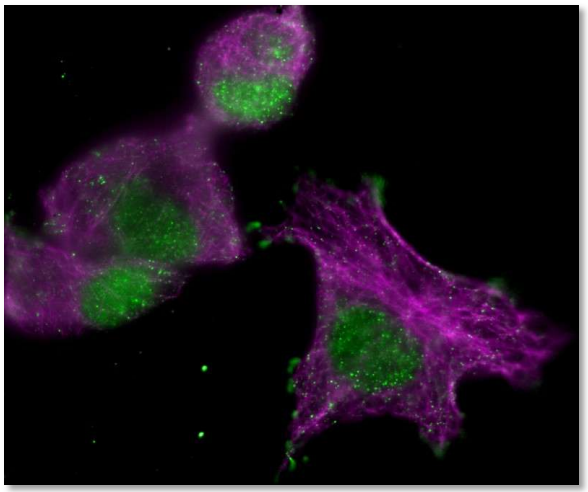
2 dimensions

The number of dimensions is the number of things you need to know to find each pixel

To find a pixel, you need: x coordinate
y coordinate (xy)

Slide courtesy: P. Bankhead

Images can have many dimensions



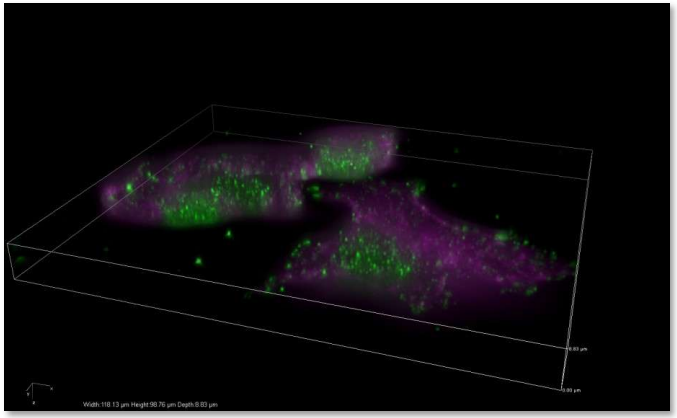
3 dimensions

The number of dimensions is the number of things you need to know to find each pixel

To find a pixel, you need: x coordinate
y coordinate Colour channel (xyc, xyt or xyz)

Slide courtesy: P. Bankhead

Images can have many dimensions



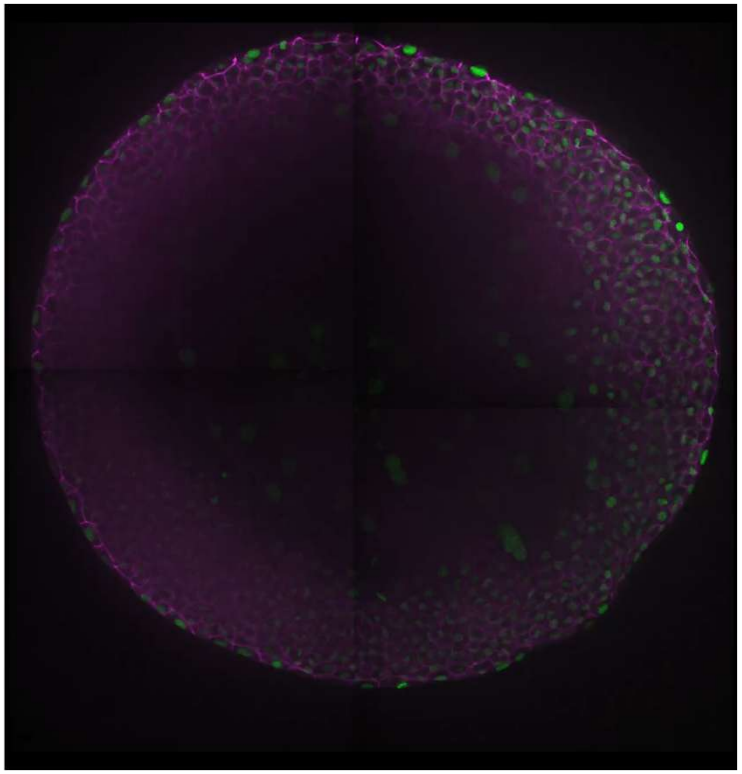
4 dimensions

The number of dimensions is the number of things you need to know to find each pixel

To find a pixel, you need: x coordinate
y coordinate Colour channel z plane
(xyzc, xycz, xyzt or xytc)

Slide courtesy: P. Bankhead

Images can have many dimensions



5 dimensions

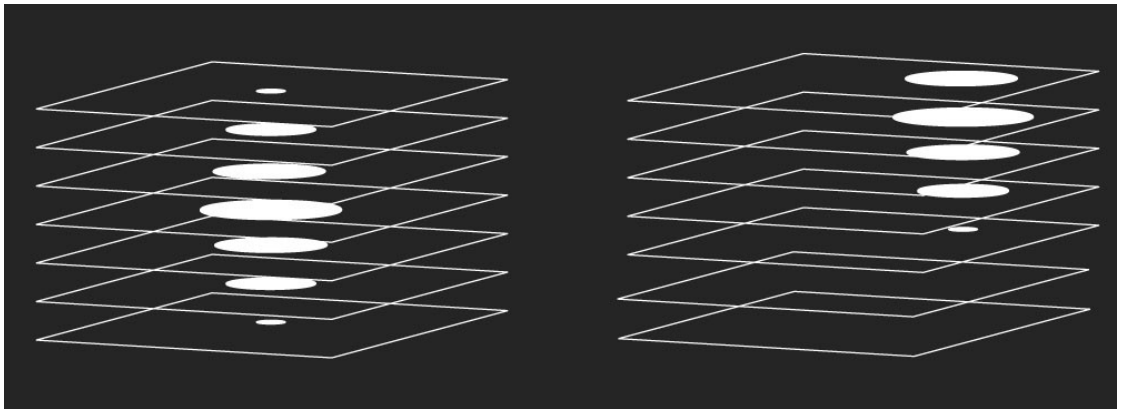
The number of dimensions is the number of things you need to know to find each pixel

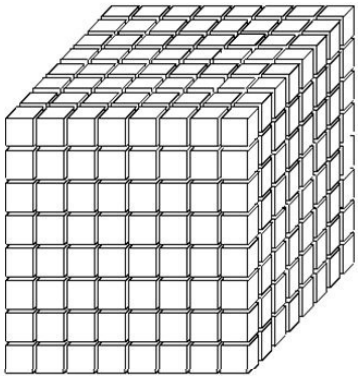
To find a pixel, you need:

x coordinate
y coordinate
Colour channel
z plane
Time point
(xyczt or other ordering)

Slide courtesy: P. Bankhead

Images can have many dimensions







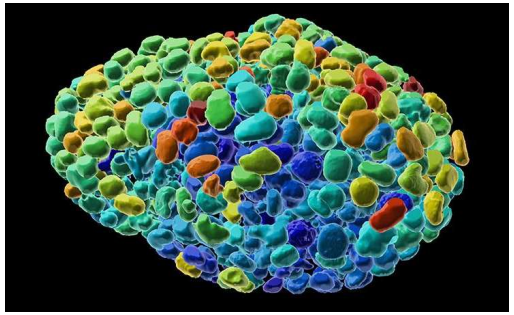
(from image to) Stack: a set of related images of the same size and bit depth (xyz or xyt).
Slices: the images that make up a stack
(from pixel to) Voxel: volumetric image element

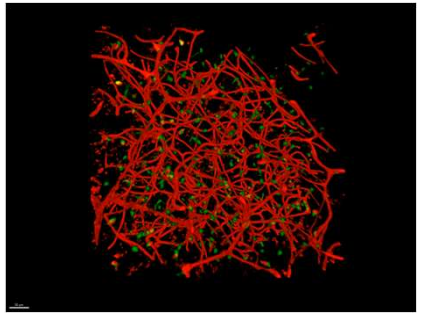
Hyperstack: When more than 3 dimensions are combined into a stack, i.e. four (4D) or five (5D) dimensions: x (width), y (height), z (slices), c (channels or wavelengths) and t (time frames).

In Fiji, hyperstacks are commonly displayed in a window with up to three scrollbars.

Commercial Software for N-D image visualisation

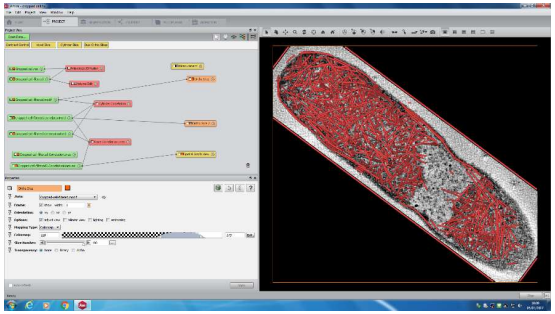


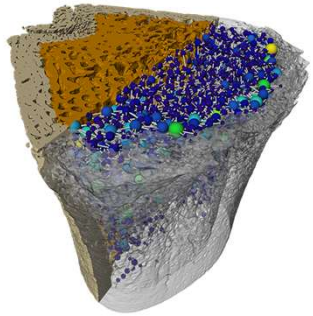




Imaris (Oxford Instruments)
https://imaris.oxinst.com/

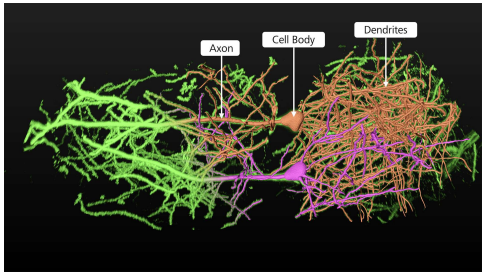
Avizo®

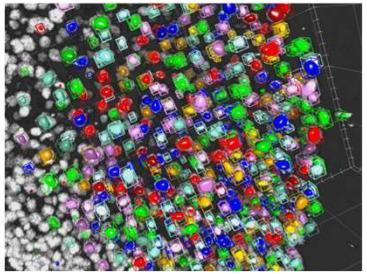




Avizo/Amira
https://www.thermofisher.com/

arivis

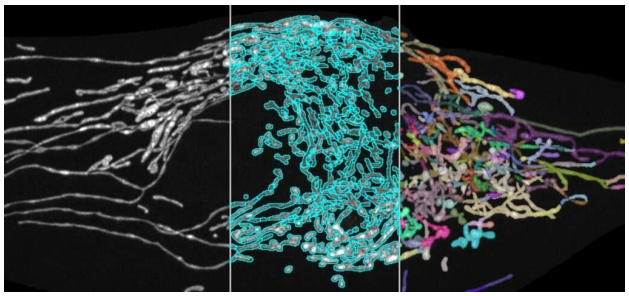




Arivis
https://www.zeiss.com/

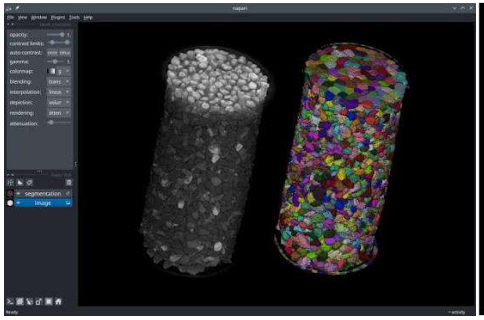
Open source Software for N-D image visualisation

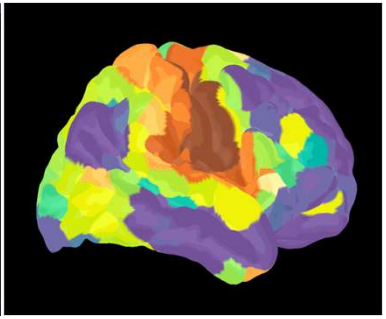




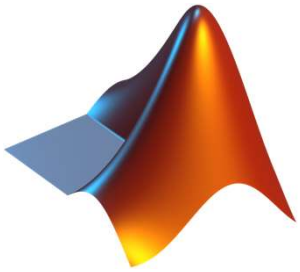
Fiji Is Just ImageJ
https://fiji.sc

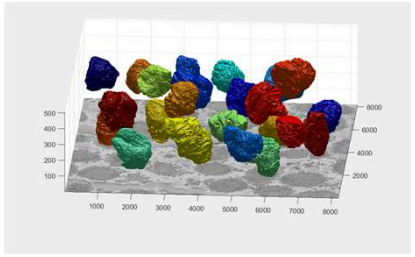


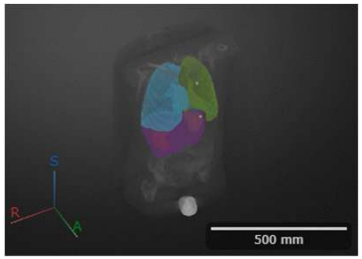




Napari
https://napari.org/stable/



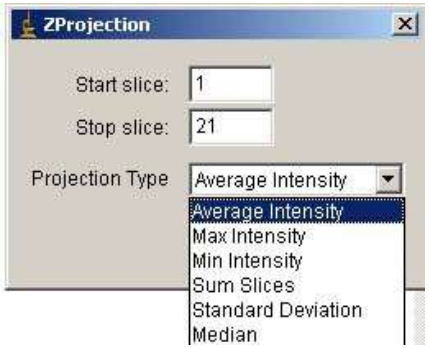




Matlab
https://www.mathworks.com/

Exercise 1 – Dimension reduction

1. (from this session material folder)
   [File > Open > moving_dot.tif]
2. **[Image > Stacks > Z Project]**



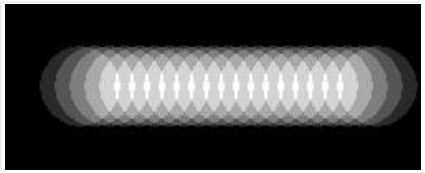


K. Miura

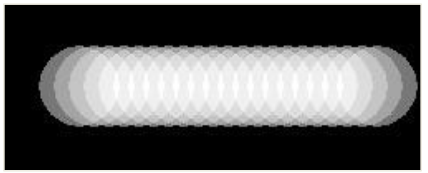
?



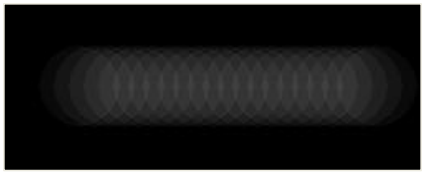
?



?



?



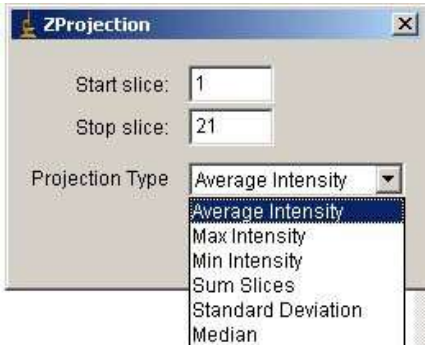
?



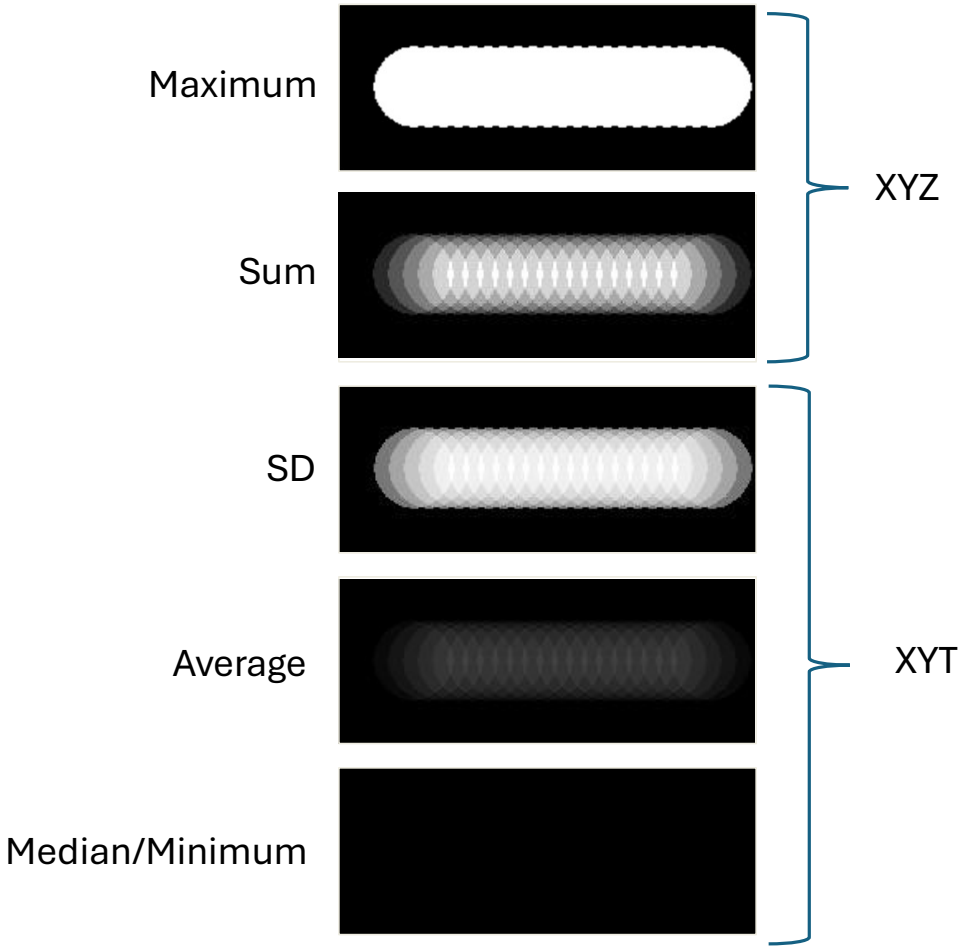
http://imagej.net/Z-functions

Exercise 1 – Dimension reduction

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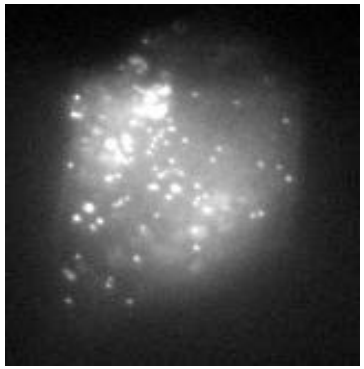


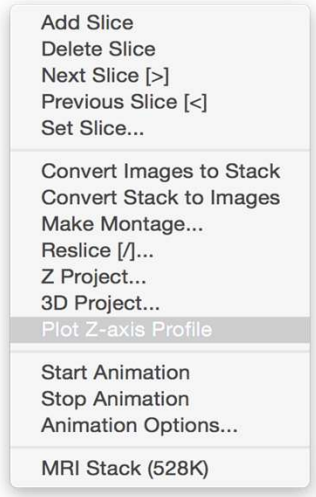


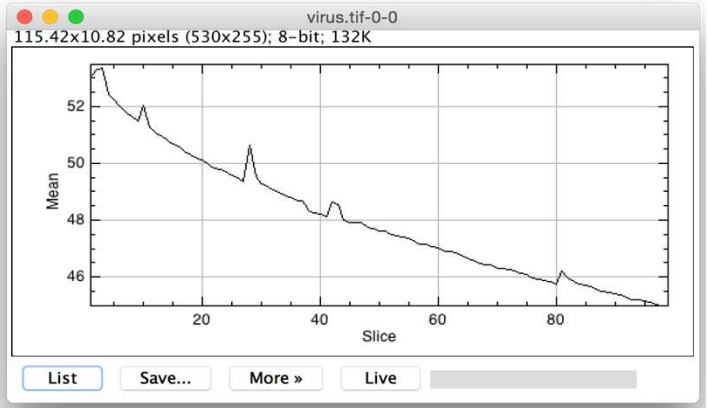
http://imagej.net/Z-functions

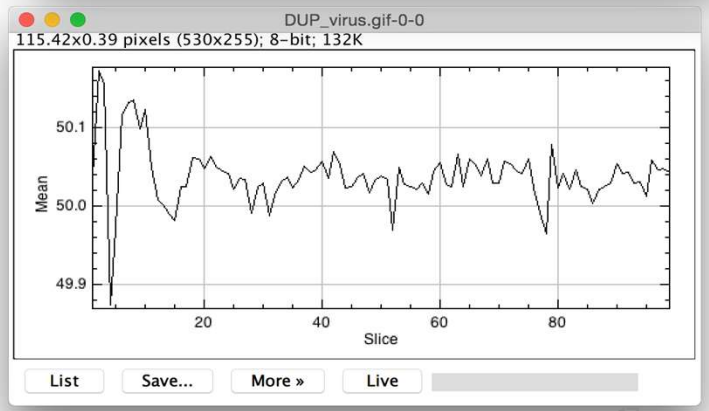
Exercise 2 – Dimension reduction

1. Open virus.tif
2. [Image > Stacks > Plot Z-axis Profile]







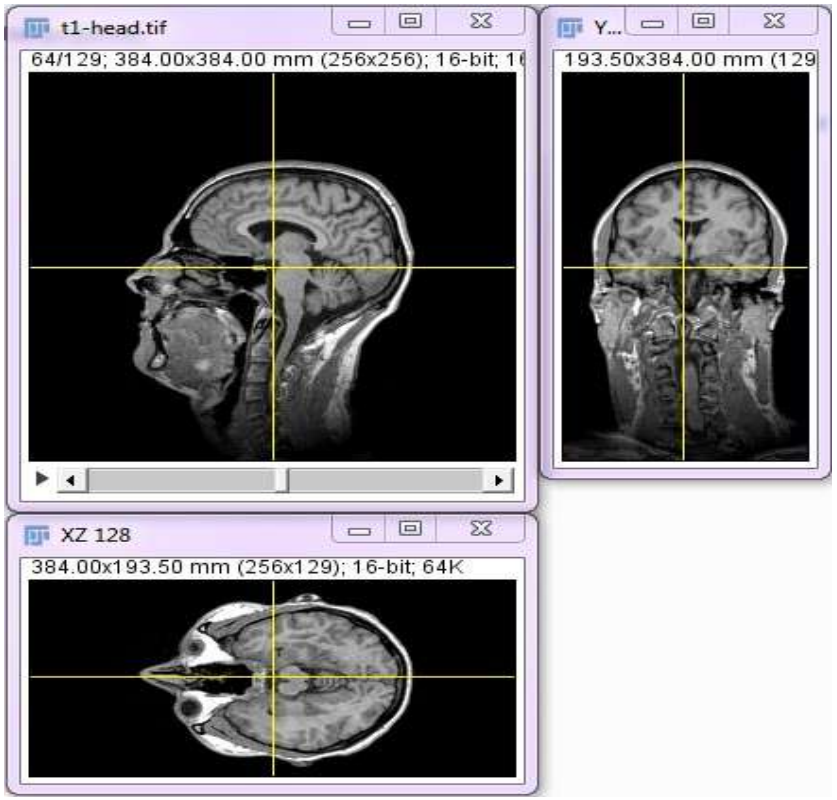


Slide courtesy: K. Miura

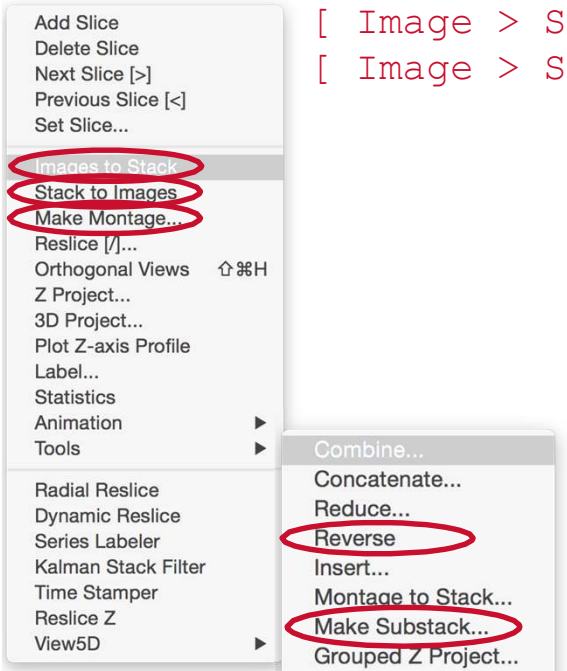
Exercise 3 – Dimension reduction

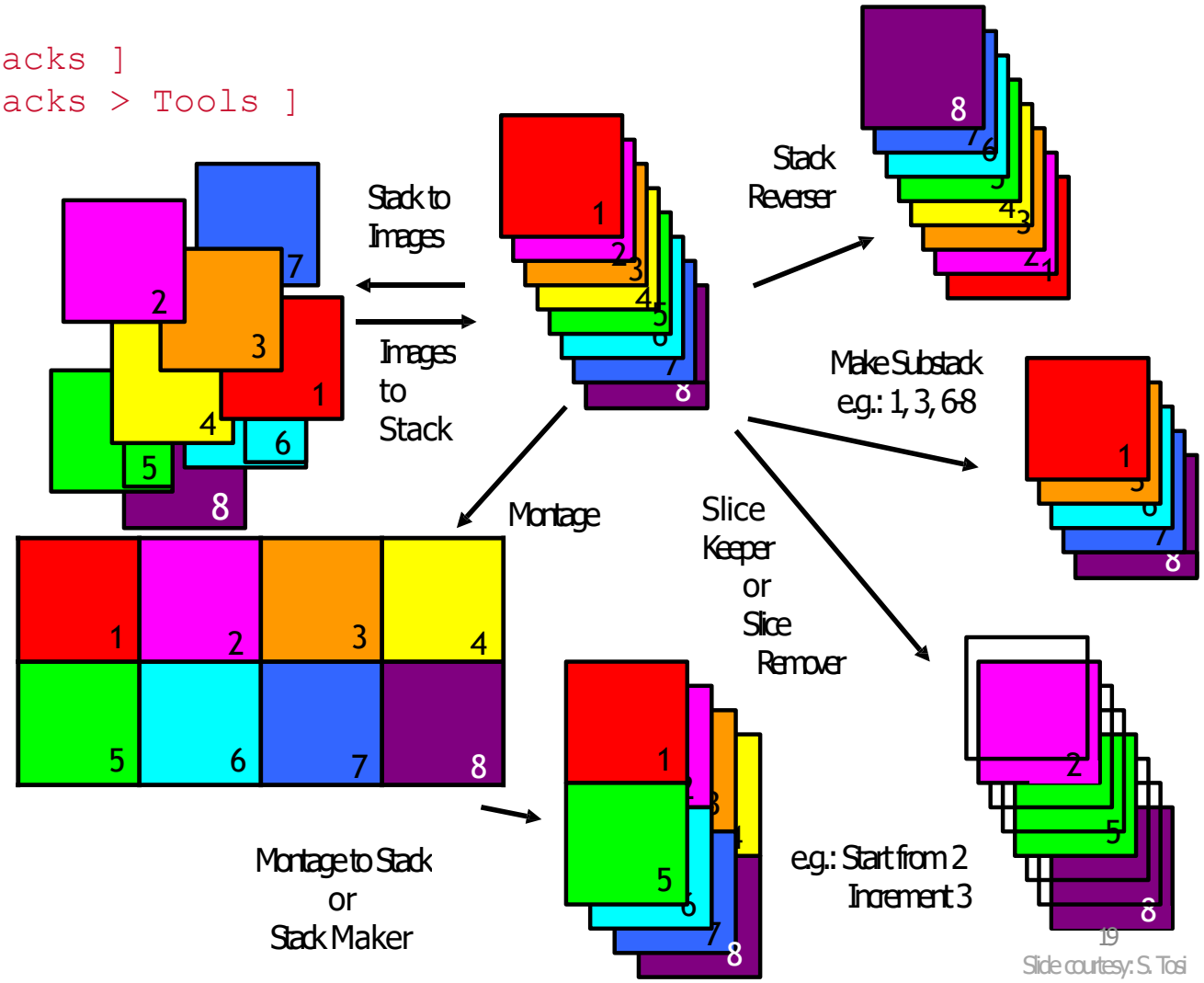
1. [File > Open Samples > T1 Head]
2. [Image > Stacks > Orthogonal views]

3. Open Plugins > Volume Viewer
4. Change Rendering Mode to Volume to visualise the head in 3D



[Image > Stacks]
[Image > Stacks > Tools]





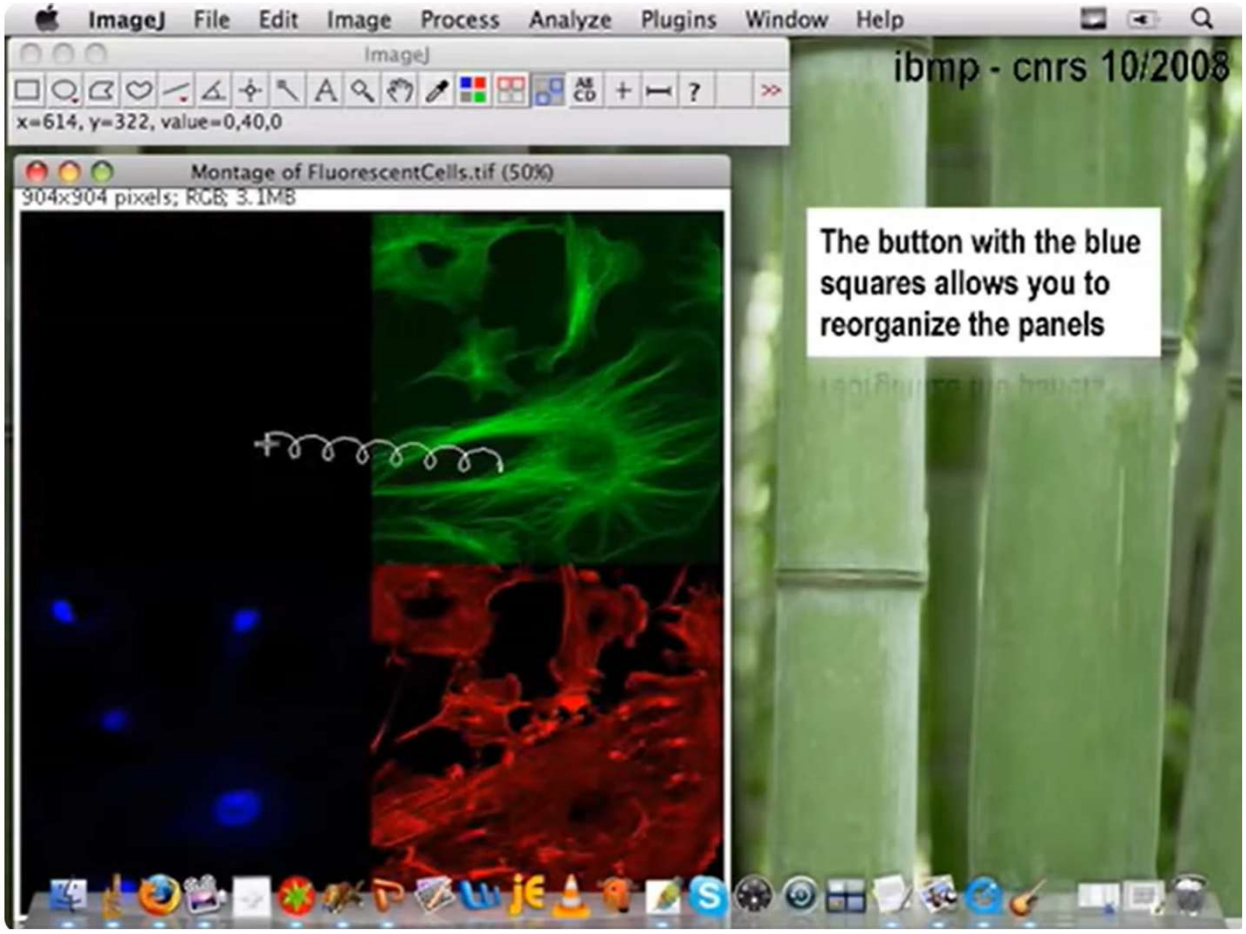
http://imagej.net/Stack-slice_Manipulations

Slide courtesy: S. Tosi

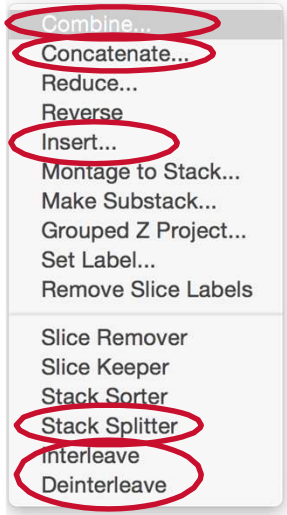
Magic Montage

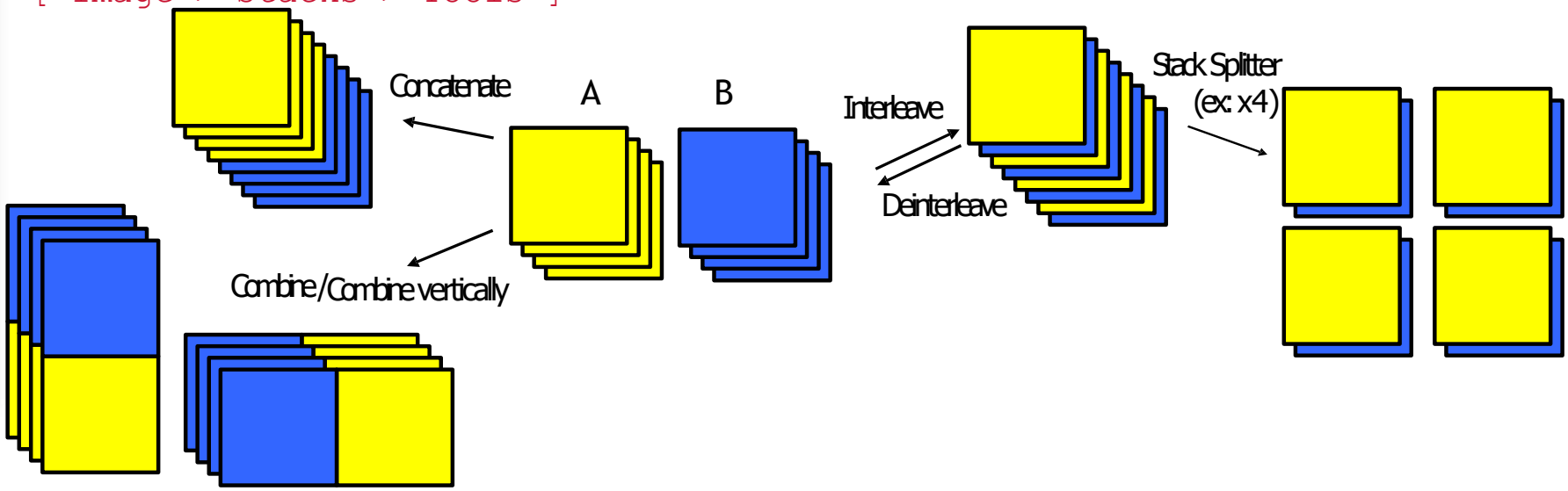
[Image > Stacks > Tools > Magic Montage Tools]

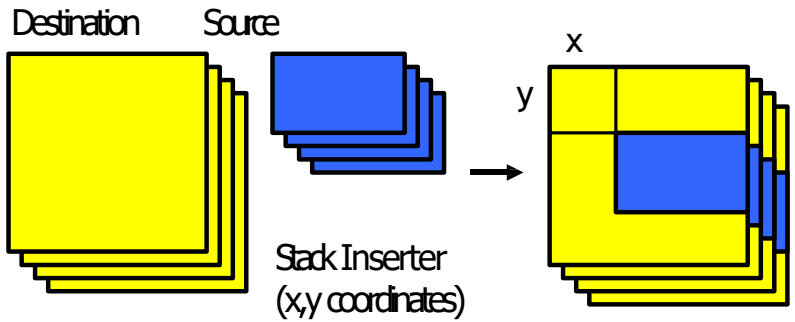
https://www.youtube.com/watch?v=y0X4olhXenk



[Image > Stacks]
[Image > Stacks > Tools]





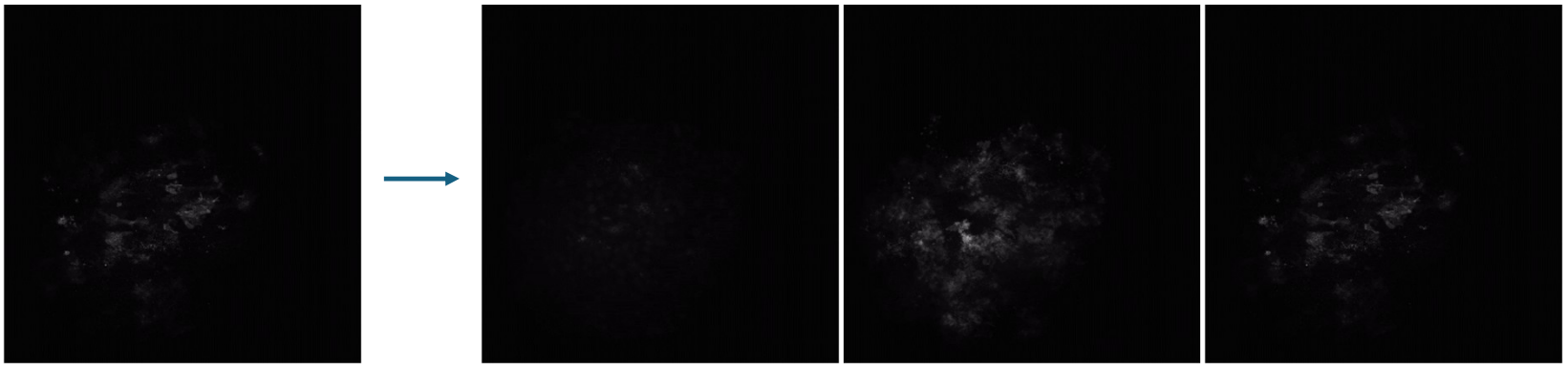


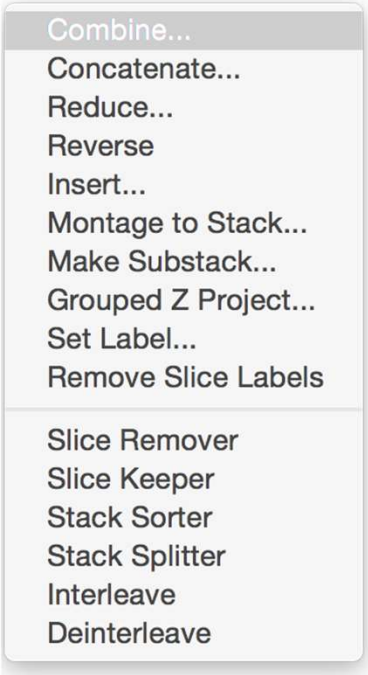
http://imagej.net/Stack-slice_Manipulations

Slide courtesy: S. Tosi

Exercise 3 – Stack manipulation

1. [File > Open > spheroid01.ome.tif]
2. Identify the issue with the stack
3. Using Image > Stack > Tools, find two ways to fix the stack and separate the channels

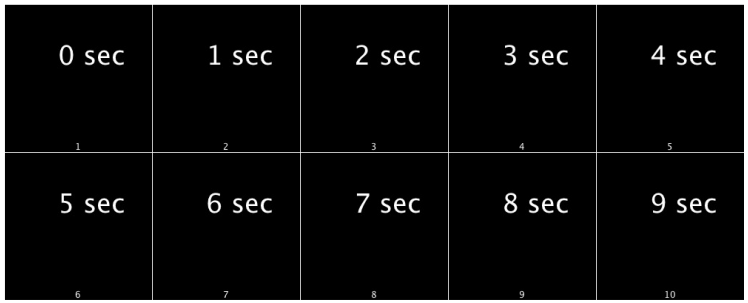




1. Image > Stack > Tools > Deinterleave (3 channels)
2. Image > Stack > Tools > Slice Keeper

Exercise 4 – Stack manipulation

1. **[File > New > Image...]**
   - **Type: 8-bit**
   - **Fill with:Black**
   - **Width: 200**
   - **Height: 200**
   - **Slices: 10**
2. **[Image > Stacks > Time Stamper]**
   - **Play with X/Y location & font size (70, 90, 30)**
3. **[Image > Stacks > Make Montage...]**
   - **Columns 5**
   - **Rows 2**
   - **Border Width 1**
   - **Label Slices On**



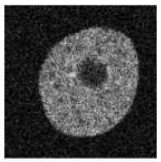
4. **Duplicate the stack,**
   - **Delete every second frame**



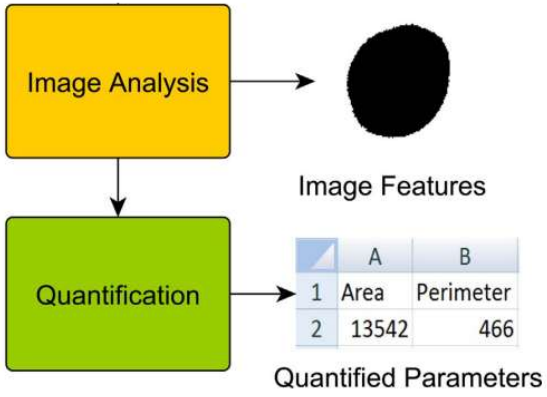
5. **Duplicate the stack, Invert the stack order and combine them side by side to make a single stack.**

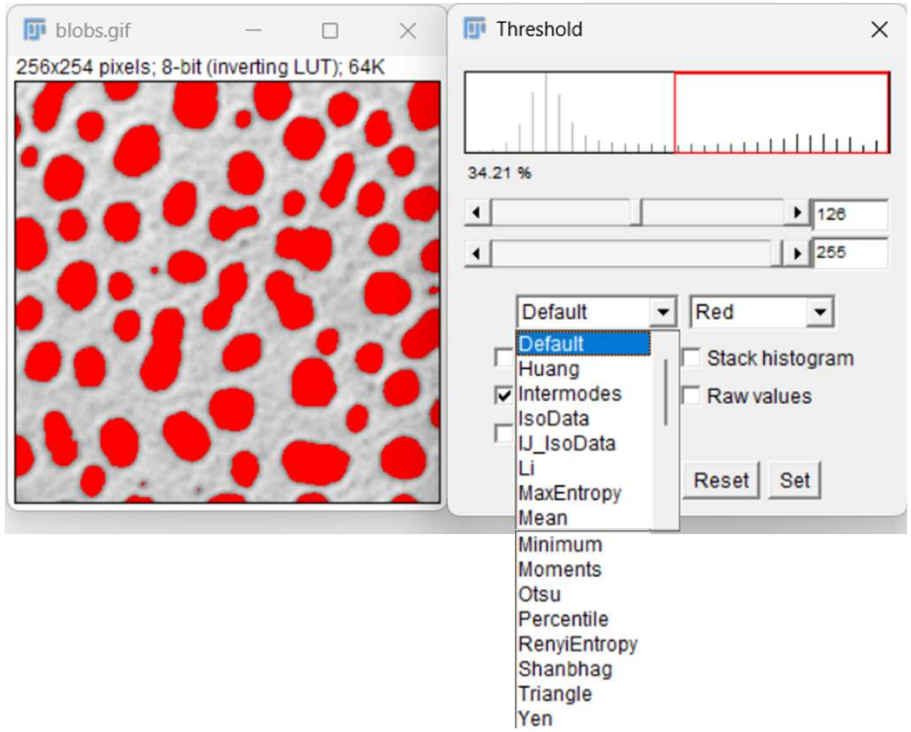


Filters and Thresholding

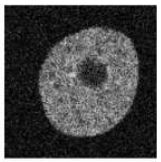


Raw Images

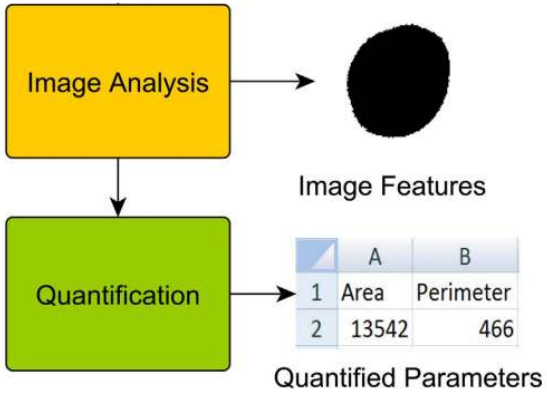




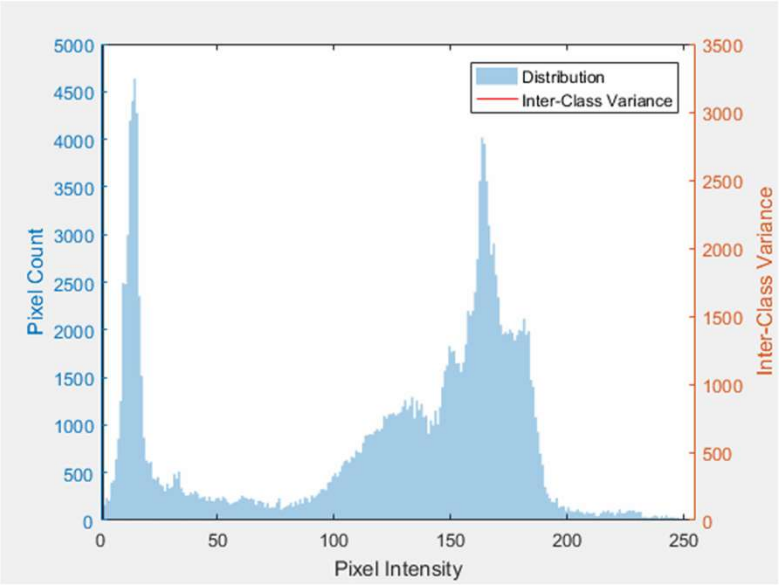
Filters and Thresholding



Raw Images



OTSU

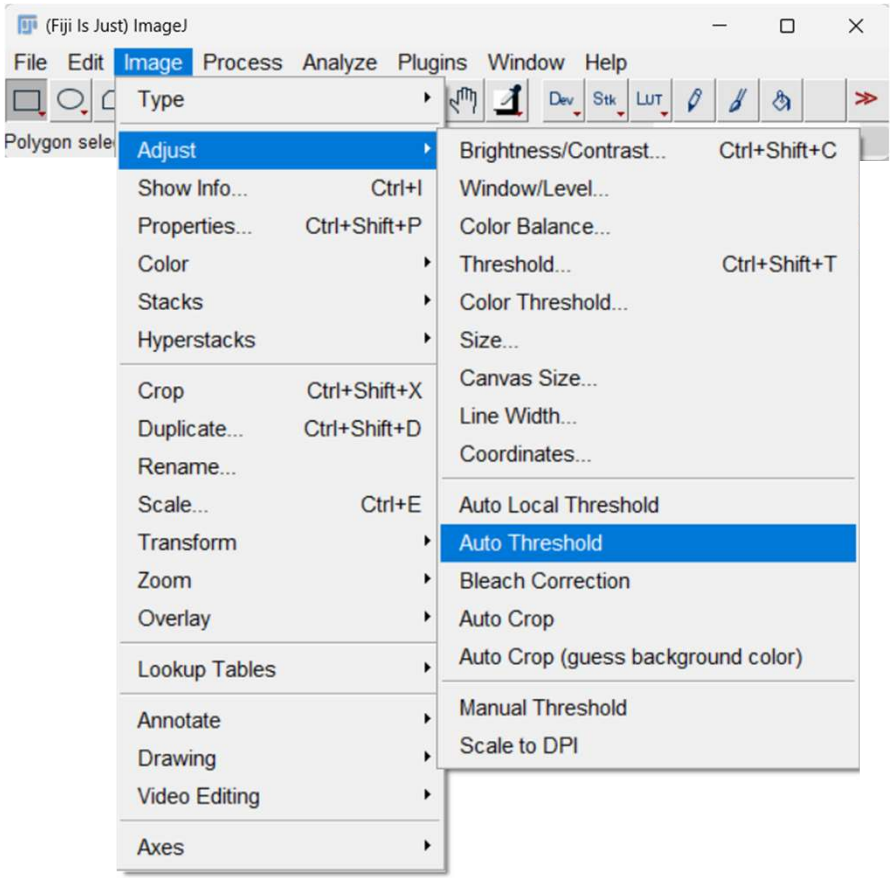


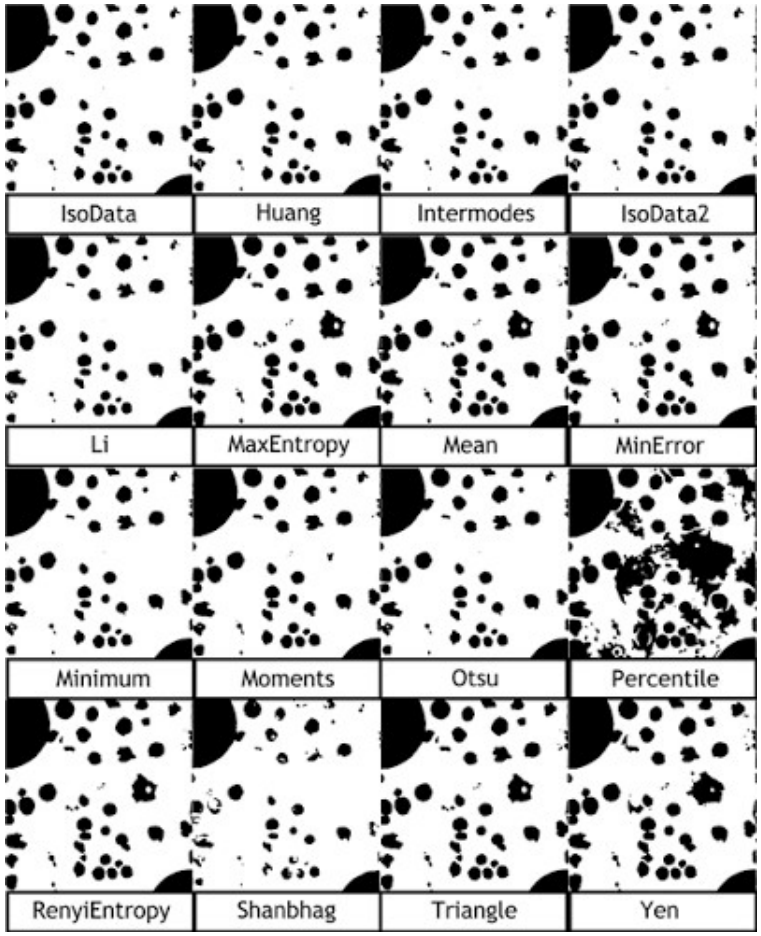
ISODATA / DEFAULT

How the IsoData Threshold Algorithm Works

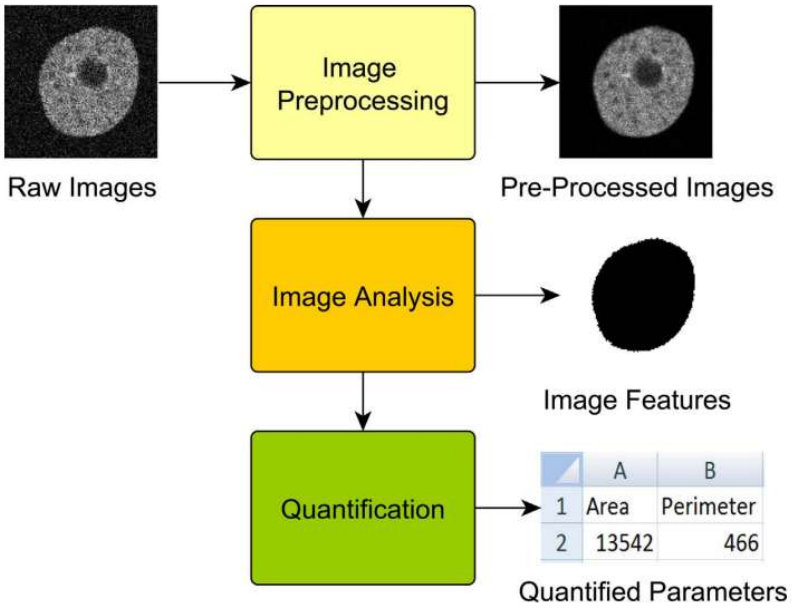
- **Initial Guess:** The histogram is split into two parts using a starting threshold—typically half of the maximum dynamic range (e.g., intensity value 127 for 8-bit images). The Generic Mapping Tools
- **Calculate Means:** The sample mean of the foreground pixels (m_H, above the threshold) and the background pixels (m_L, at or below the threshold) are computed. www.fmwconcepts.com +1
- **Update Threshold:** A new threshold is calculated as the average of these two means: $T = (m_L + m_H)/2$. Universiteit van Amster... +1
- **Iterate:** The process repeats iteratively with the updated threshold value until successive iterations yield no change in T. www.fmwconcepts.com

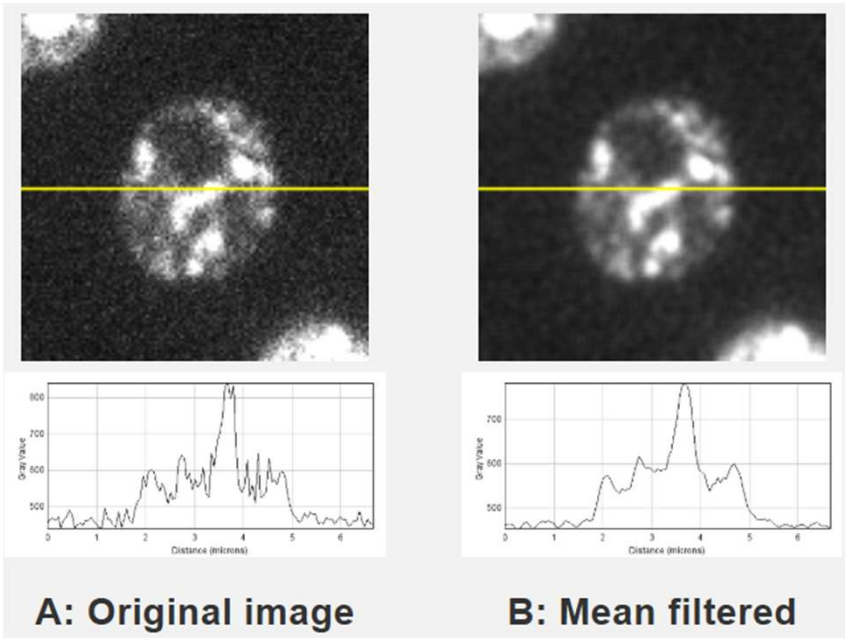
Filters and Thresholding

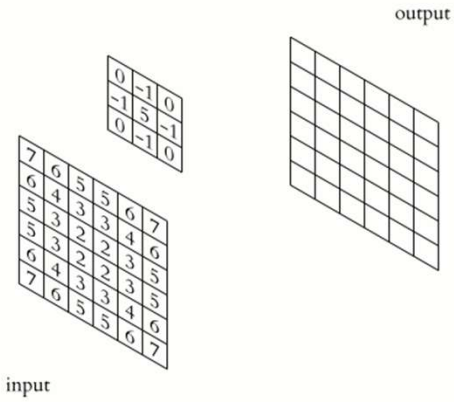




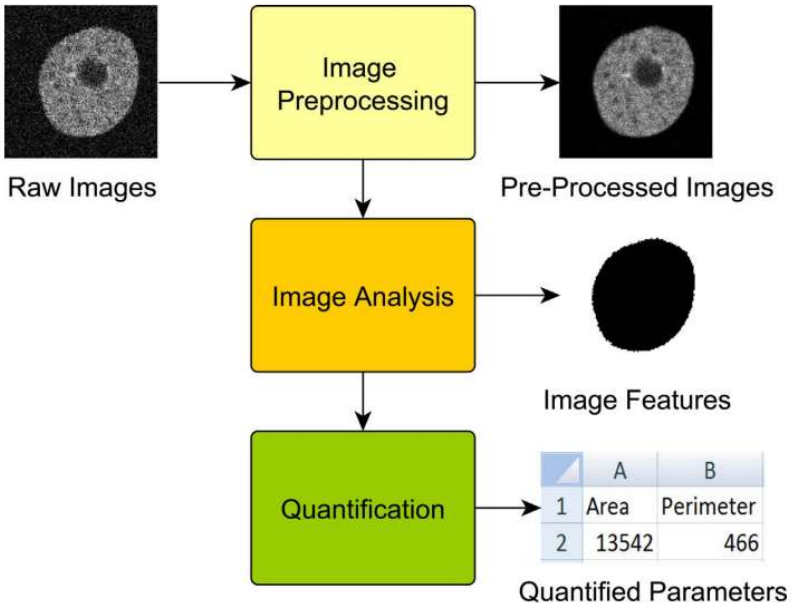
Filters and Thresholding

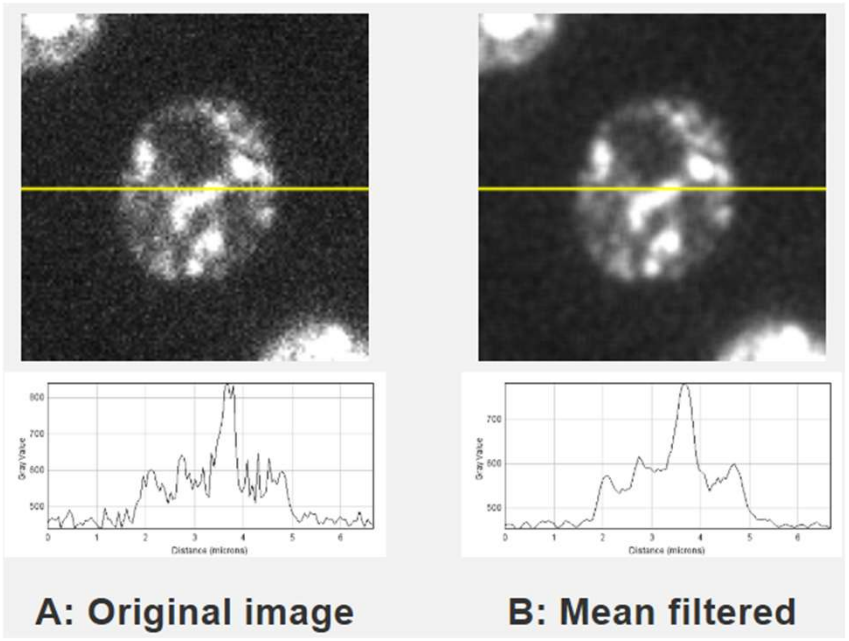






Filters and Thresholding





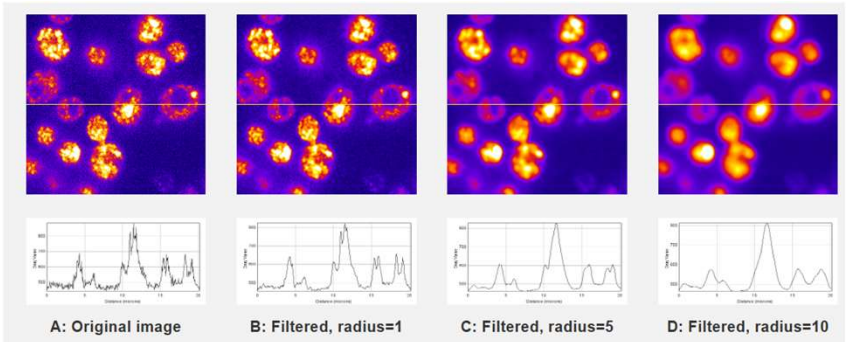
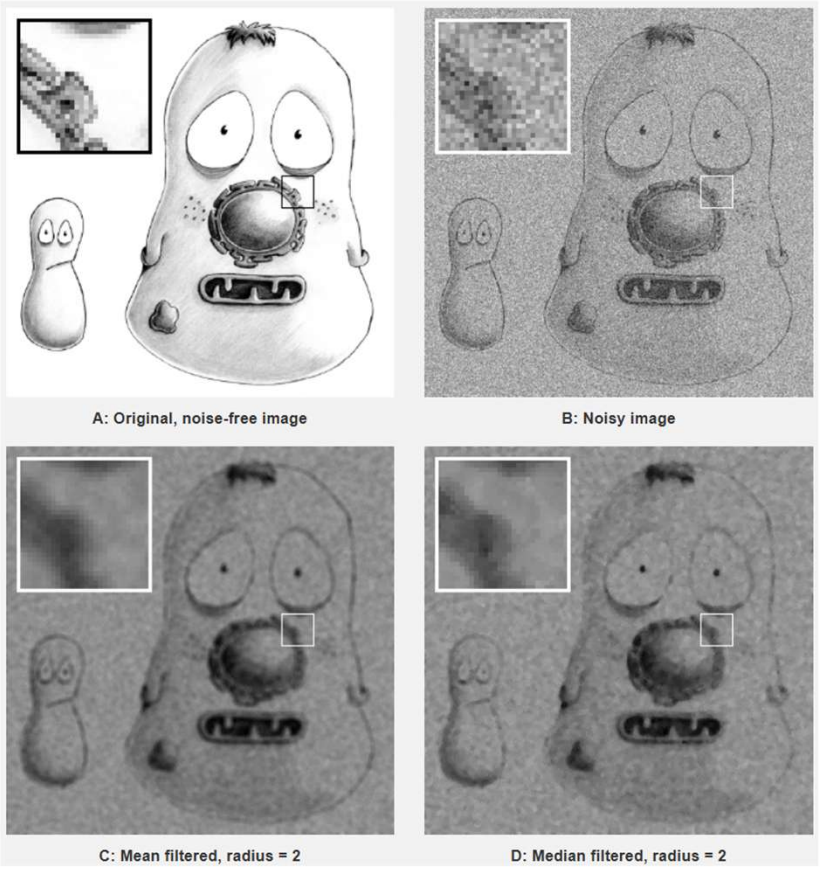


Figure 2: Smoothing an image using mean filters with different radii.

Filters and Thresholding



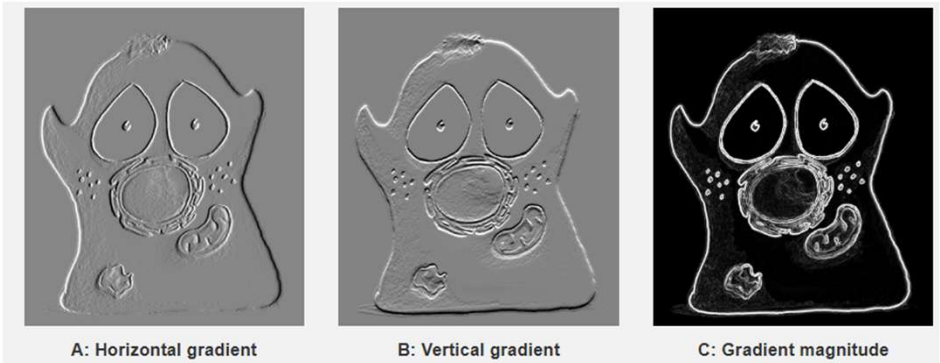
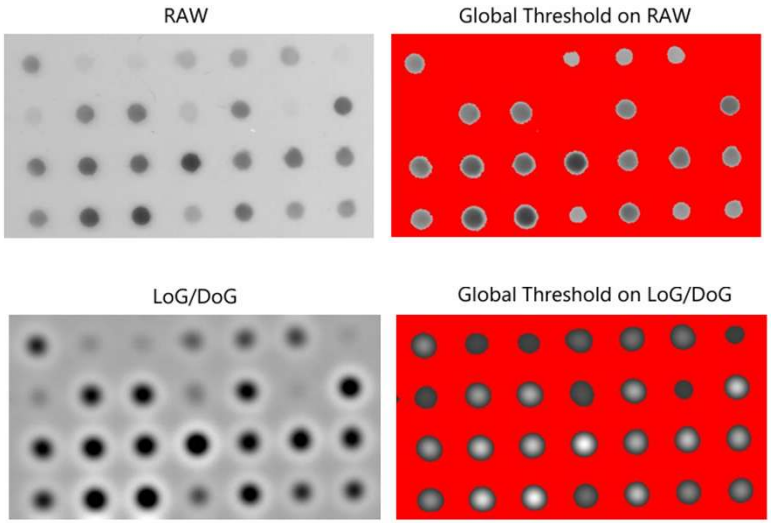
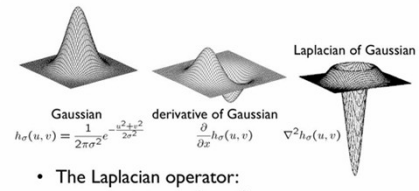


Figure 5: Using gradient filters and the gradient magnitude for edge enhancement.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$





- The Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Slide credit: K. Grauman

Exercise 5 – Whole analysis pipeline

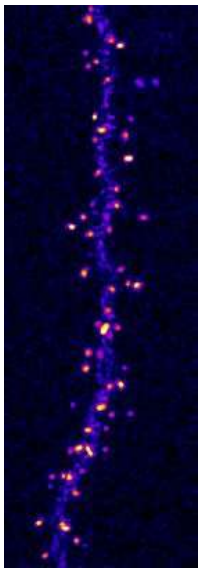


Image courtesy of:
Dr. Sebastien Tosi
Danish Bio Imaging IACF

Image Preprocessing

1. [Open > GFP neurons spines.tif]
2. [Plugins > FeatureJ > FeatureJ Laplacian]
   - apply the filter with a smoothing scale of 2

Image Analysis

3. On the filtered image: [Image > Adjust > Threshold]
   - Untick Dark background, i.e. you see red spots on the image where the neurons are, then click Apply
   - Choose convert to mask
   - un-tick "calculate threshold for each image"
   - apply the threshold
4. [Process > Binary > Fill holes]
5. [Analyze > 3D OC Options]
   - tick the interested measurements: e.g. Volume, Nb of Obj. voxels, centroid, Integrated density, Mean gray value,
   - tick "show masked image" & select the original stack in the "redirect to" drop down menu.
   - untick "show numbers".

Quantification

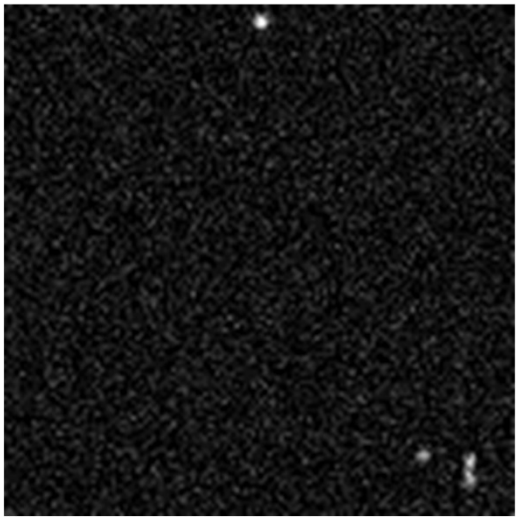
6. [Analyze > 3D objects counter]
   - threshold: 128; minimum size filter: 5 voxels
   - tick objects, statistics and summary
7. ([Analyze > Tools > Synchronize Windows])
   - Explore objects map stack (1 color per object) together with the original image

http://imagescience.org/meijering/software/featurej/

Slide courtesy: M. Bosch

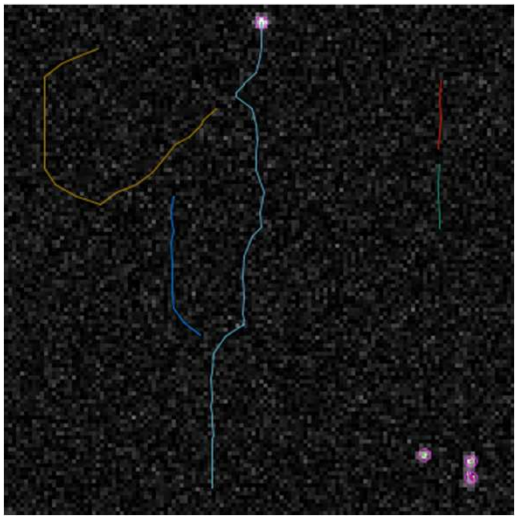
Questions ?

Particles/Objects tracking



Key parameters for good tracking:

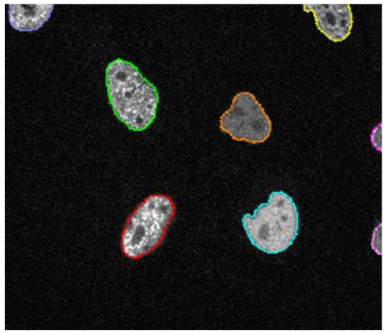
- Segmentation :
  - Good S/N
  - Reduce bleaching (toxicity)
- Tracking :
  - Adapted acquisition rate
  - Complexity of the tracking (merging, splitting, blinking)



Typical outputs:

- Speed
- Number of split/merge events
- Total distance travelled
- Persistency of the movement

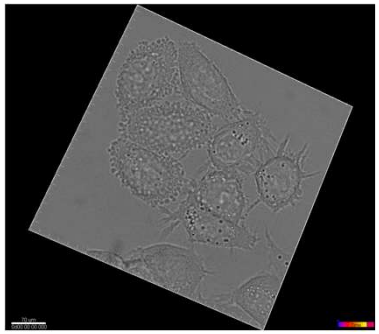
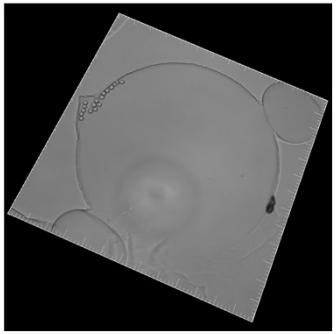
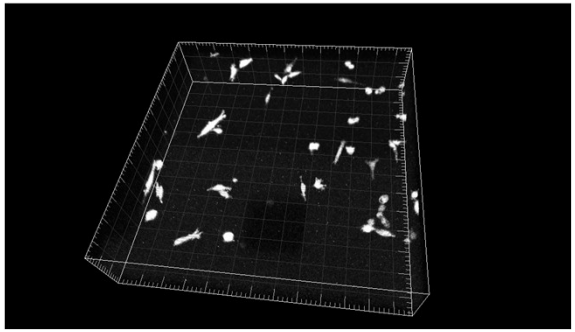
Particles/Objects tracking



"Segmenting and tracking moving cells in time-lapse sequences is a challenging task"
https://celltrackingchallenge.net/ (Since 2012, last edition in Athens 27th May 2024)

3 benchmarks :

- Cell segmentation
- Cell tracking
- Cell linking

Segmentation

	Generalizability	BF-C2DL-HSC*	BF-C2DL-MuSC*	DIC-C2DH-HeLa*	Fluo-C2DL-Huh7	Fluo-C2DL-MSC*	Fluo-C3DH-A549*	Fluo-C3DH-H157*	Fluo-C3DL-MDA231*	Fluo-N2DH-GOWT1*	Fluo-N2DL-HeLa*	Fluo-N3DH-CE*	Fluo-N3DH-CHO*	Fluo-N3DL-DRO	Fluo-N3DL-TRIC	Fluo-N3DL-TRIF	PhC-C2DH-U373*	PhC-C2DL-PSC*	Fluo-C3DH-A549-SIM	Fluo-N2DH-SIM+	Fluo-N3DH-SIM+
OP_{CSB}	0.849	0.926	0.883	0.926	0.885	0.761	0.954	0.938	0.810	0.952	0.957	0.847	0.926	0.873	0.906	0.895	0.961	0.859	0.977	0.905	0.949
	0.842	0.908	0.879	0.912	0.879	0.760	0.952	0.930	0.807	0.948	0.956	0.842	0.923	0.863	0.891	0.886	0.958	0.859	0.975	0.905	0.899
	0.799	0.908	0.876	0.912	0.859	0.746	0.936	0.927	0.771	0.948	0.954	0.830	0.913	0.827	0.875	0.885	0.956	0.857	0.943	0.904	0.866
SEG	0.791	0.855	0.782	0.877	0.811	0.687	0.908	0.899	0.710	0.938	0.923	0.759	0.925	0.760	0.821	0.793	0.931	0.756	0.955	0.832	0.906
	0.761	0.826	0.778	0.871	0.791	0.655	0.903	0.889	0.710	0.933	0.922	0.729	0.917	0.738	0.818	0.776	0.929	0.743	0.951	0.830	0.820
	0.731	0.826	0.774	0.866	0.782	0.649	0.878	0.888	0.704	0.931	0.922	0.722	0.914	0.713	0.782	0.775	0.927	0.740	0.886	0.829	0.759
DET	0.922	0.997	0.983	0.975	0.968	0.876	1.000	0.988	0.911	0.980	0.994	0.990	0.954	0.988	0.994	0.997	0.990	0.975	1.000	0.981	0.992
	0.906	0.995	0.982	0.961	0.959	0.851	1.000	0.982	0.907	0.976	0.992	0.981	0.945	0.985	0.994	0.997	0.990	0.975	1.000	0.981	0.978
	0.871	0.994	0.980	0.960	0.956	0.832	1.000	0.978	0.904	0.970	0.992	0.979	0.934	0.970	0.986	0.994	0.988	0.972	1.000	0.980	0.974

AC	BFR-GE	BGU-IL	CALT-US	CSU-CN	CUNI-CZ	DESU-US	DKFZ-GE	FR-GE
HD-GE (BMCV)	IGFL-FR	JAN-US	KIT-GE	KTH-SE	LEID-NL	MON-AU	MPI-GE (CBG)	MSU-RU
MU-CZ	MU-US	ND-US	OX-UK	PURD-US	RWTH-GE	TUG-AT	UCH-CL	UFRGS-BR
UNSW-AU	UVA-NL	WARW-UK						

Created on 2023-12-18, www.celltrackingchallenge.net

Legend: * (dataset included in or result transferred from the generalizability study)

Tracking

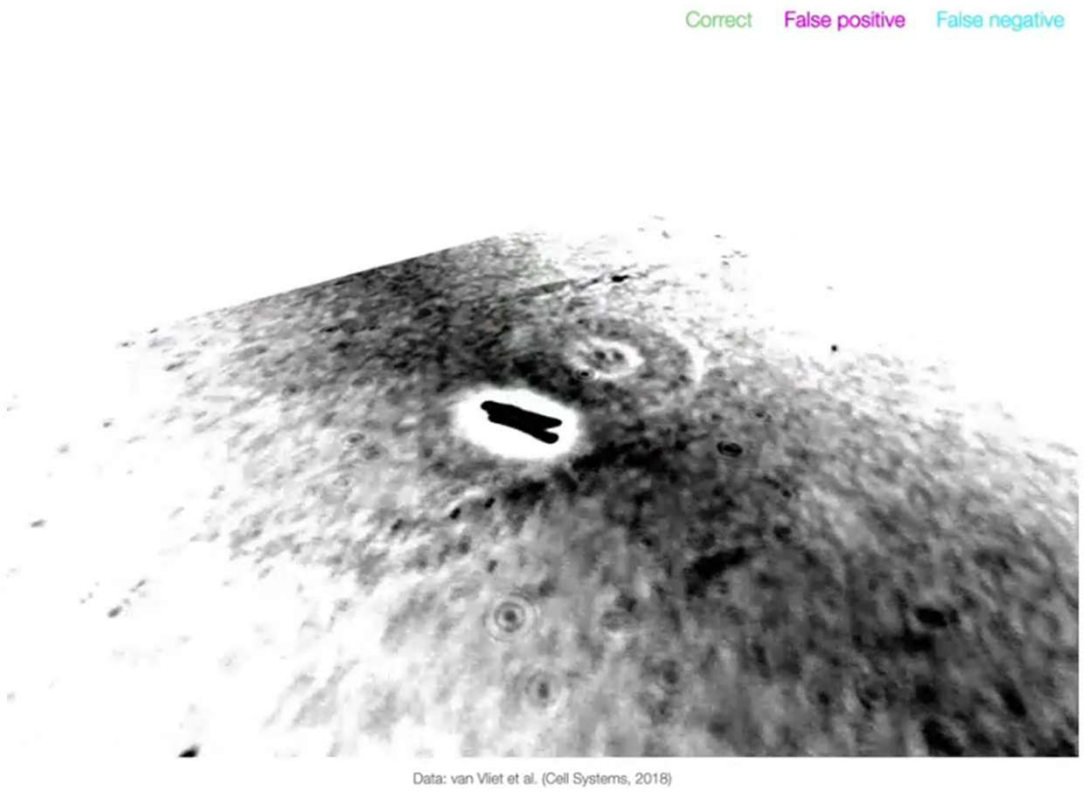
	Generalizability	BF-C2DL-HSC*	BF-C2DL-MuSC*	DIC-C2DH-HeLa*	Fluo-C2DL-Huh7	Fluo-C2DL-MSC*	Fluo-C3DH-A549*	Fluo-C3DH-H157*	Fluo-C3DL-MDA231*	Fluo-N2DH-GOWT1*	Fluo-N2DL-HeLa*	Fluo-N3DH-CE*	Fluo-N3DH-CHO*	Fluo-N3DL-DRO	Fluo-N3DL-TRIC	Fluo-N3DL-TRIF	PhC-C2DH-U373*	PhC-C2DL-PSC*	Fluo-C3DH-A549-SIM	Fluo-N2DH-SIM+	Fluo-N3DH-SIM+
OP_{CTB}	0.791	0.907	0.878	0.909	0.875	0.759	0.938	0.938	0.797	0.951	0.956	0.844	0.926	0.708	0.867	0.841	0.954	0.855	0.920	0.905	0.897
	0.732	0.906	0.875	0.904	0.843	0.740	0.931	0.929	0.761	0.941	0.953	0.829	0.912	0.617	0.816	0.804	0.951	0.854	0.913	0.903	0.865
	0.637	0.901	0.870	0.898	0.772	0.710	0.925	0.925	0.757	0.940	0.953	0.808	0.911	0.591	0.787	0.785	0.951	0.851	0.865	0.896	0.848
SEG	0.728	0.826	0.782	0.863	0.791	0.655	0.876	0.889	0.710	0.931	0.923	0.759	0.917	0.613	0.791	0.746	0.924	0.743	0.840	0.830	0.820
	0.649	0.826	0.778	0.853	0.751	0.649	0.863	0.888	0.642	0.929	0.922	0.729	0.905	0.567	0.766	0.684	0.923	0.740	0.827	0.829	0.759
	0.530	0.818	0.774	0.852	0.690	0.645	0.849	0.878	0.632	0.929	0.919	0.722	0.902	0.397	0.680	0.654	0.922	0.738	0.730	0.825	0.746
TRA	0.855	0.988	0.974	0.955	0.960	0.873	1.000	0.987	0.884	0.979	0.993	0.987	0.953	0.802	0.952	0.955	0.985	0.968	1.000	0.979	0.974
	0.815	0.987	0.971	0.954	0.934	0.839	1.000	0.980	0.882	0.976	0.992	0.979	0.948	0.785	0.942	0.936	0.982	0.967	1.000	0.978	0.972
	0.768	0.985	0.971	0.954	0.865	0.788	1.000	0.976	0.880	0.967	0.991	0.975	0.935	0.785	0.854	0.886	0.982	0.966	1.000	0.977	0.967

AC	BGU-IL	CAS-CN	CUNI-CZ	DESU-US	DREX-US	FR-GE	HIT-CN	IGFL-FR
JAN-US	KIT-GE	KTH-SE	LEID-NL	MPI-GE (CBG)	MU-CZ	MU-US	ND-US	PURD-US
RWTH-GE	TUG-AT	UVA-NL						

Created on 2023-12-18, www.celltrackingchallenge.net

Legend: * (dataset included in or result transferred from the generalizability study)

Particles/Objects tracking



Trackastra - Tracking by Association with Transformers

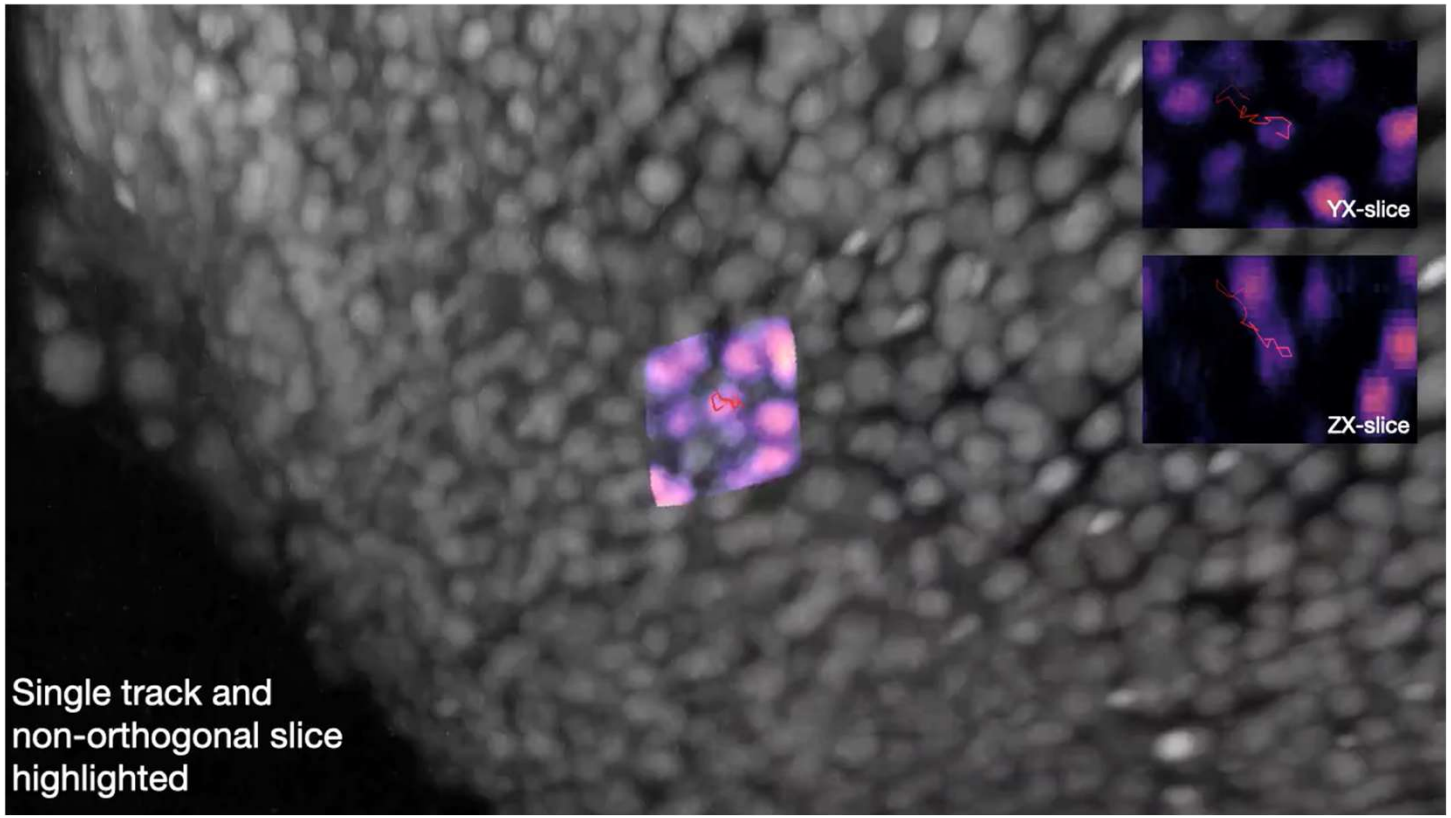
Trackastra is a cell tracking approach that links already segmented cells in a microscopy timelapse by predicting associations with a transformer model. It comes with pretrained models that perform well out of the box for many types of live-cell imaging data.

https://github.com/weigertlab/trackastra

Particles/Objects tracking

Large-scale cell tracking under segmentation uncertainty.

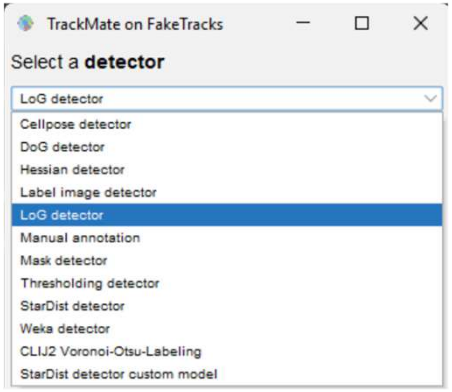
Ultrack is a versatile and scalable cell tracking method designed to address the challenges of tracking cells across 2D, 3D, and multichannel timelapse recordings, especially in complex and crowded tissues where segmentation is often ambiguous. By evaluating multiple candidate segmentations and employing temporal consistency, Ultrack ensures robust performance under segmentation uncertainty.

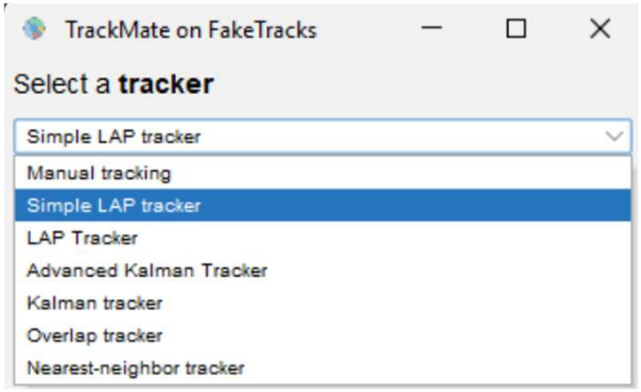


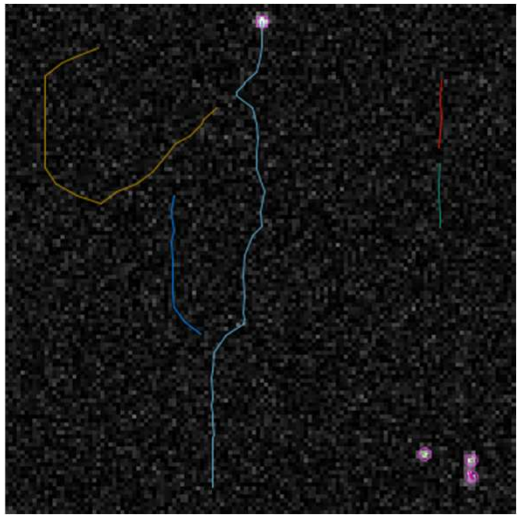
https://github.com/royerlab/ultrack

Particles/Objects tracking

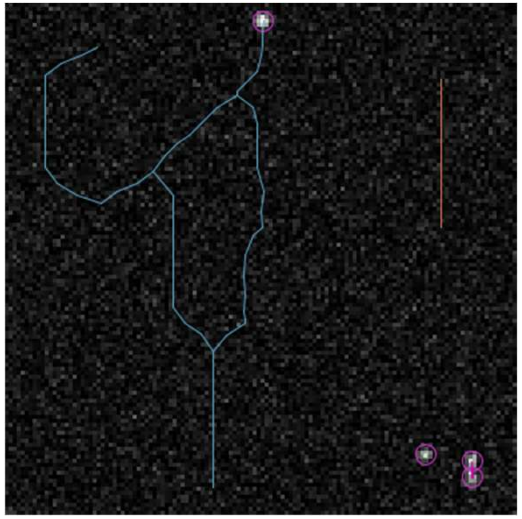




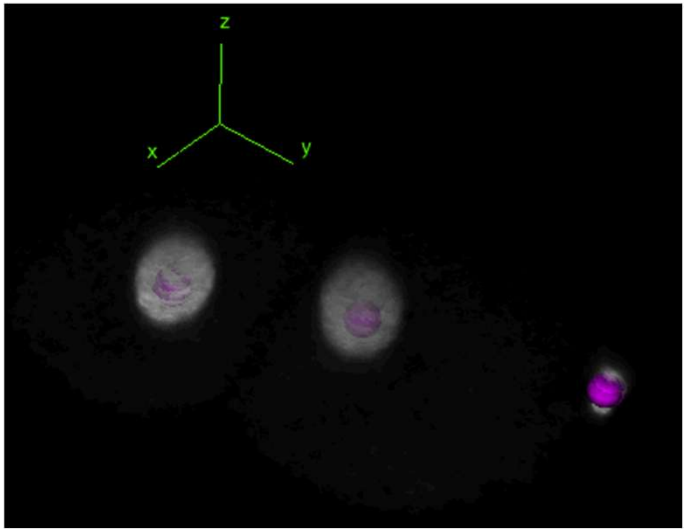




Simple LAP tracker



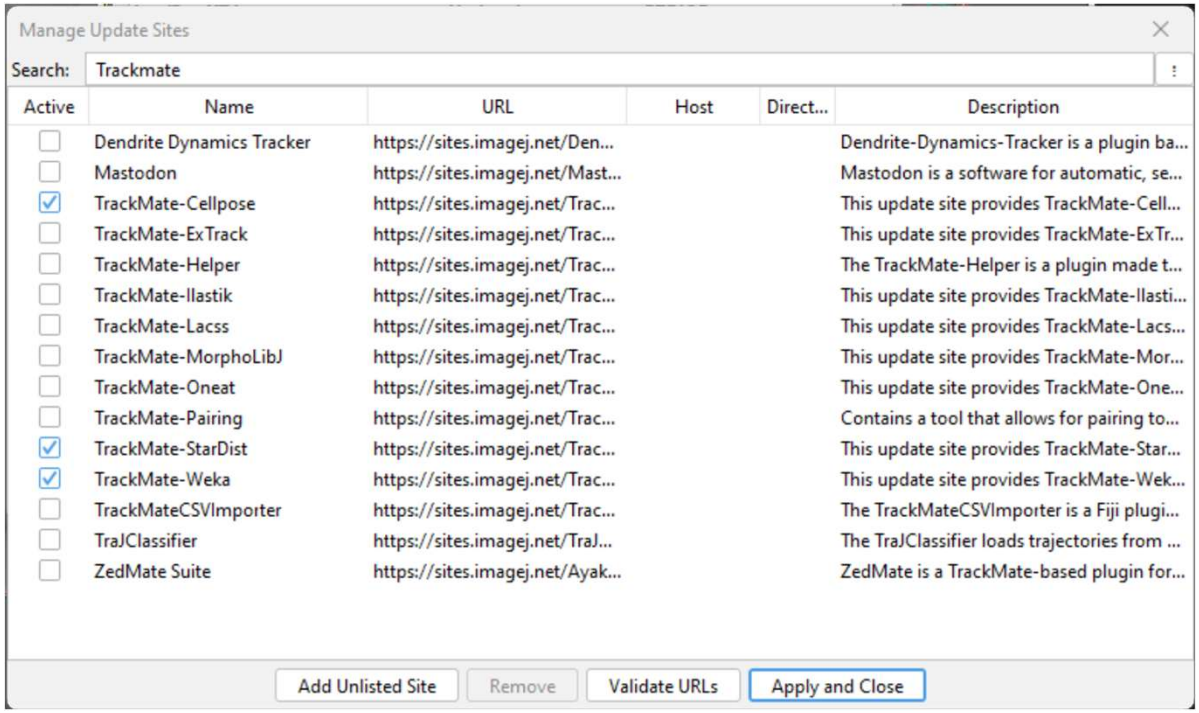
LAP tracker



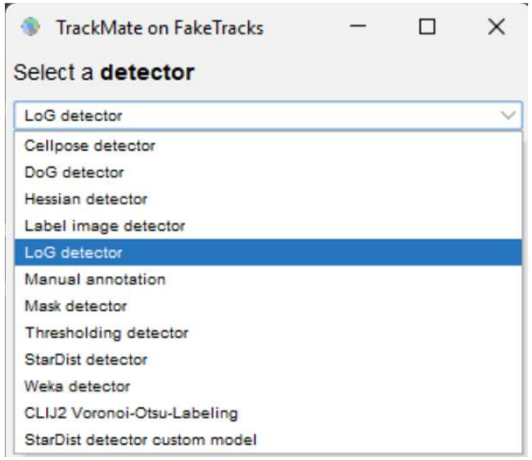
Particles/Objects tracking

How to install Trackmate in Fiji

- *Trackmate-Helper looks cool*



Particles/Objects tracking – Blob Detector



LoG – Laplacian of Gaussian

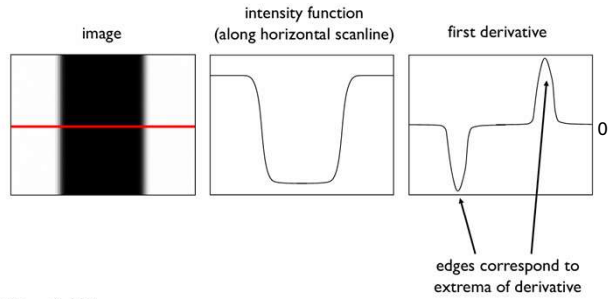
This detector applies a LoG (Laplacian of Gaussian) filter to the image, with a sigma suited to the blob estimated size. Calculations are made in the Fourier space. The maxima in the filtered image are searched for, and maxima too close from each other are suppressed. A quadratic fitting scheme allows to do sub-pixel localization.

DoG – Difference of Gaussian

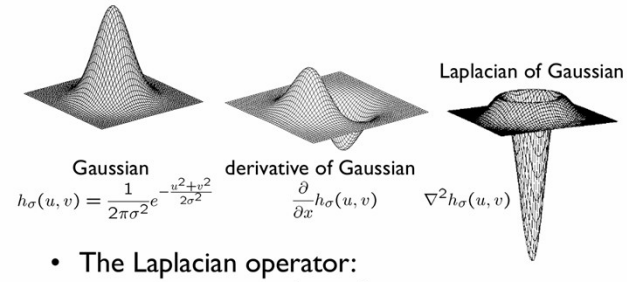
This detector is based on an approximation of the LoG operator by differences of Gaussian (DoG). Computations are made in direct space. It is the quickest for small spot sizes (~5 pixels).

Hessian

This detector is based on computing the determinant of the Hessian matrix of the image to detector bright blobs.
It can be configured with a different spots size in XY and Z. It can also return a normalized quality value, scaled from 0 to 1 for the spots of each time-point.
As discussed in Mikolajczyk et al.(2005), this detector has a better edge response elimination than the LoG detector and is suitable for detect spots in images with many strong edges.



Slide credit: K. Grauman

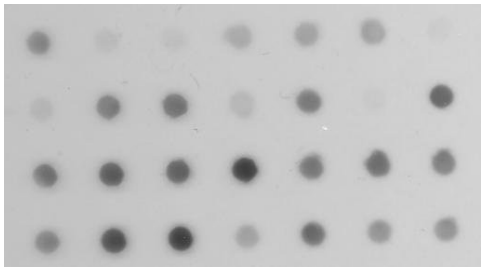


- The Laplacian operator:

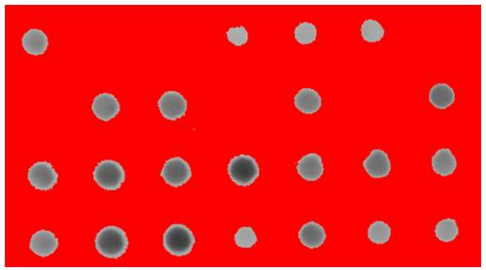
$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Slide credit: K. Grauman

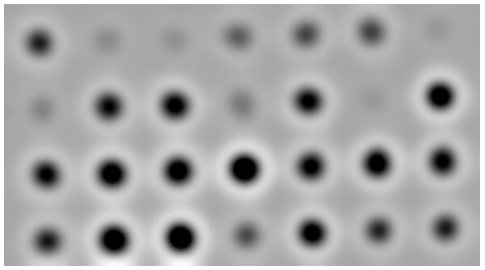
RAW



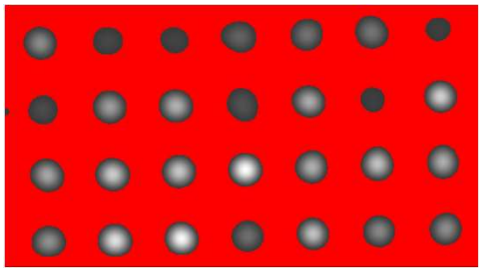
Global Threshold on RAW



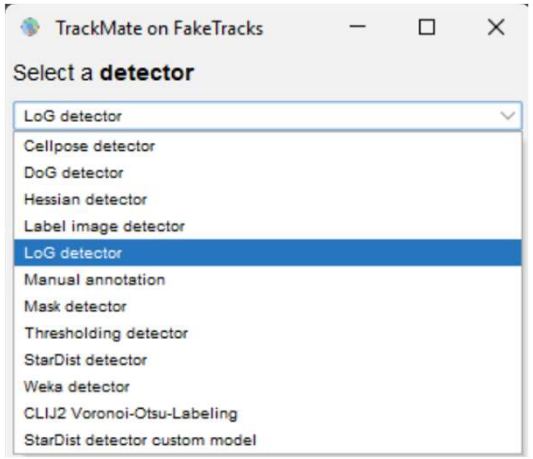
LoG/DoG

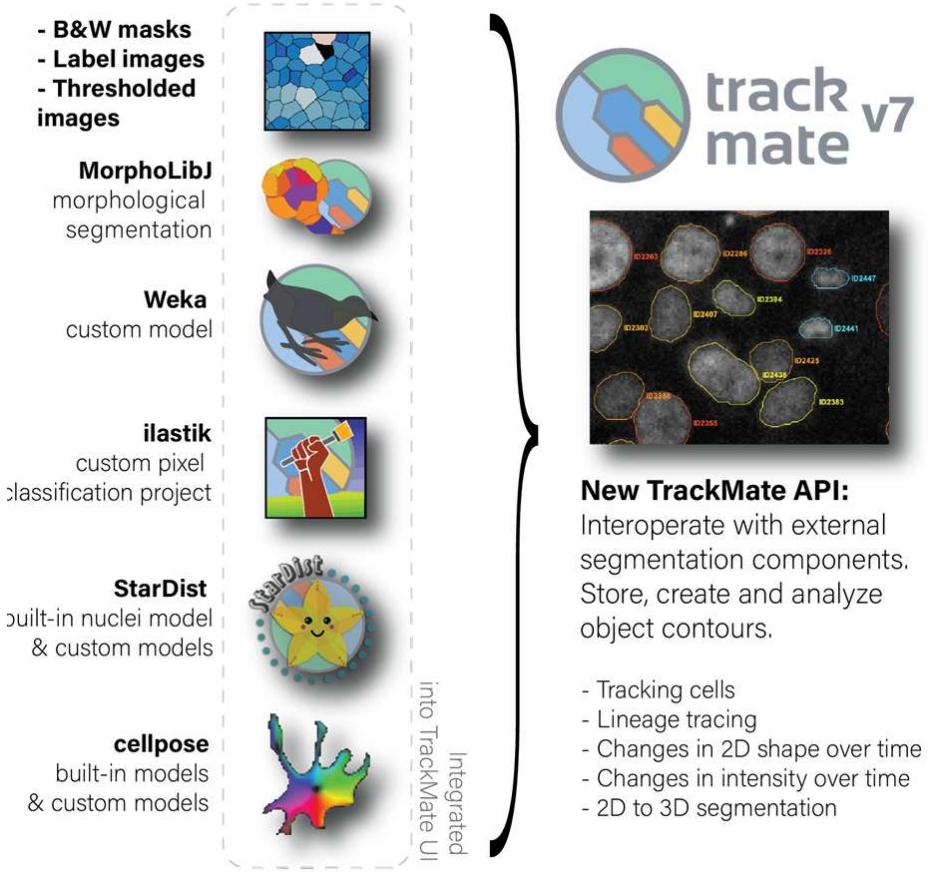


Global Threshold on LoG/DoG



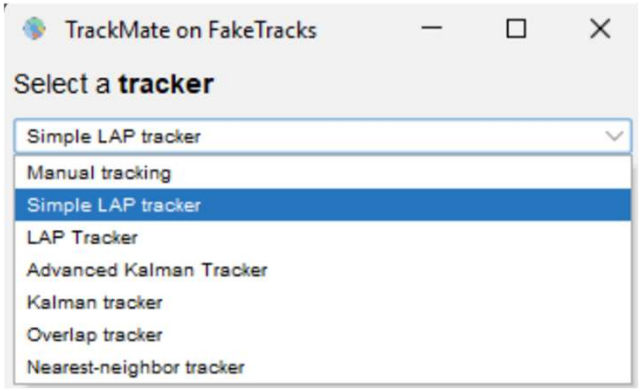
Particles/Objects tracking – ML/DL Detector





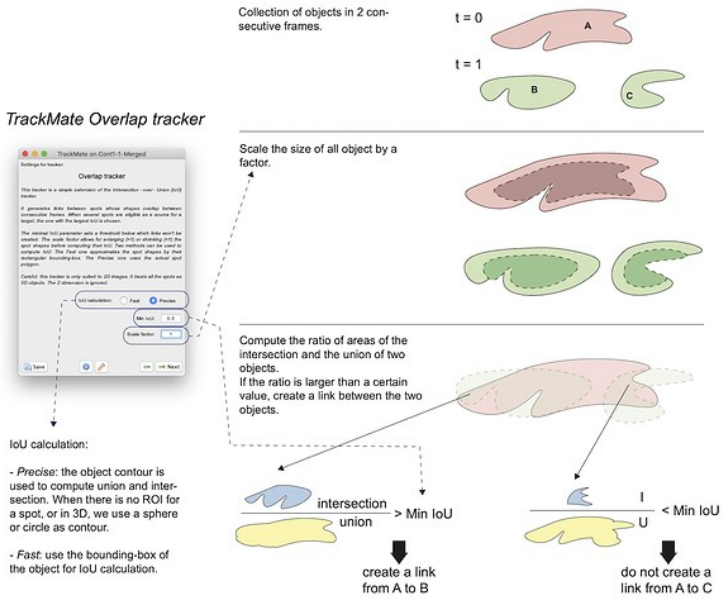
Ershov, D., Phan, MS., Pylvänäinen, J.W. *et al.* TrackMate 7: integrating state-of-the-art segmentation algorithms into tracking pipelines. *Nat Methods* **19**, 829–832 (2022)

Particles/Objects tracking - Trackers

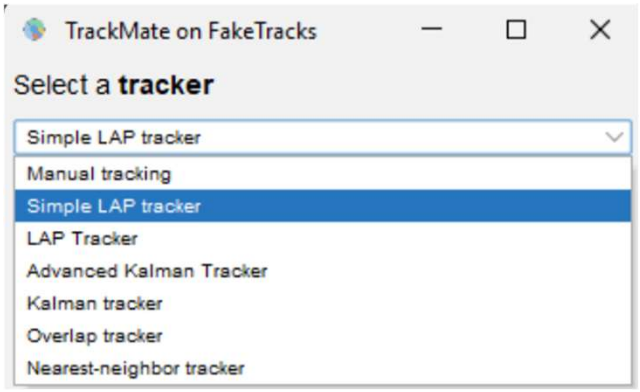


Overlap - a tracker based on overlapping object contours, suitable for object with complex shapes and limited movements.

Nearest-Neighbor – the simplest tracker.



Particles/Objects tracking - Trackers

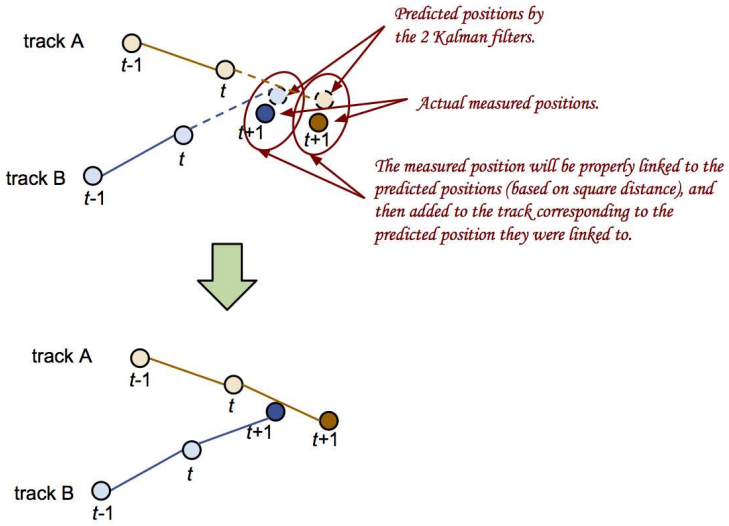


Kalman - a tracking algorithm suitable for objects that move with a nearly constant velocity.

The linear motion tracker relies on the Kalman filter to *predict* the most probable position of a particle undergoing constant velocity movement.
Tracks are initiated from the first two frames, using the classical LAP framework with the Jaqaman cost matrix (see above), using the square distance as cost. The user can set what is the maximal distance allowed for the initial search with the **Initial search radius** setting.

LAP - a tracking algorithms based on the Linear Assignment Problem algorithm

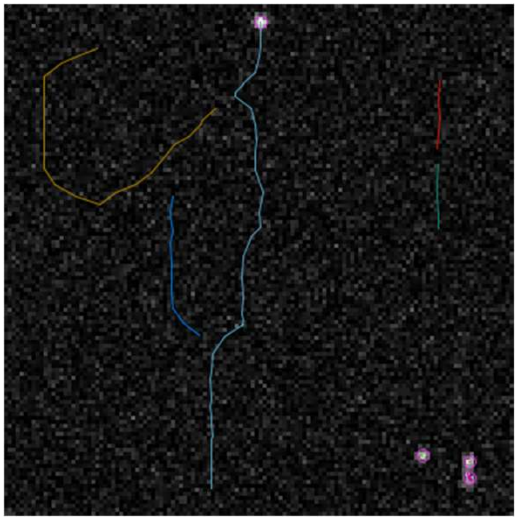
All the linking costs for these two trackers are based on the particle-to-particle square distance. If this tracker had to be summarized in one sentence, it would be the following: **The Simple LAP tracker and the LAP tracker are well suited for particle undergoing Brownian motion.** Of course, they will be fine for a non-Brownian motion as long as the particles are not too dense. In the paper, costs depend solely on the spot-to-spot distance, possibly weighted by the difference in spot intensity. Here, we offer to the user to tune costs by adding penalties on spot features



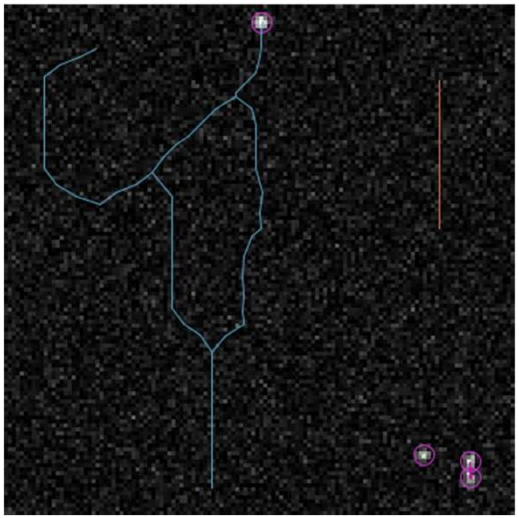
Jaqaman, K (2008). Robust single-particle tracking in live-cell time-lapse sequences. *Nature Methods*, *5*(8), 695–702. doi:10.1038/nmeth.1237

Exercise 1

Using File > Image Samples > Track for Trackmates
Try to get those 2 tracks overlays



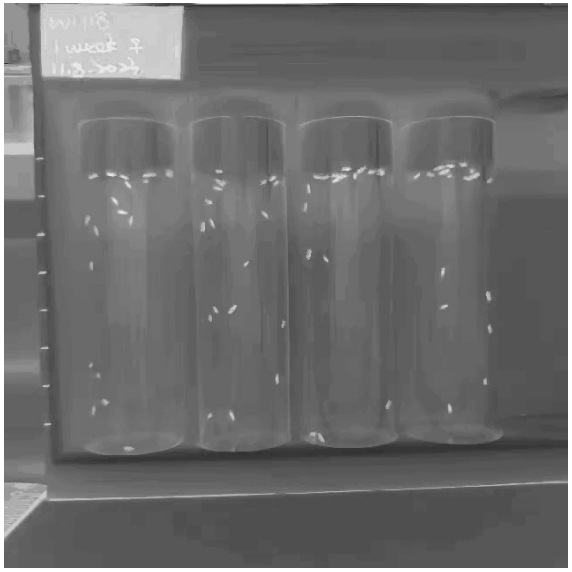
Simple LAP tracker

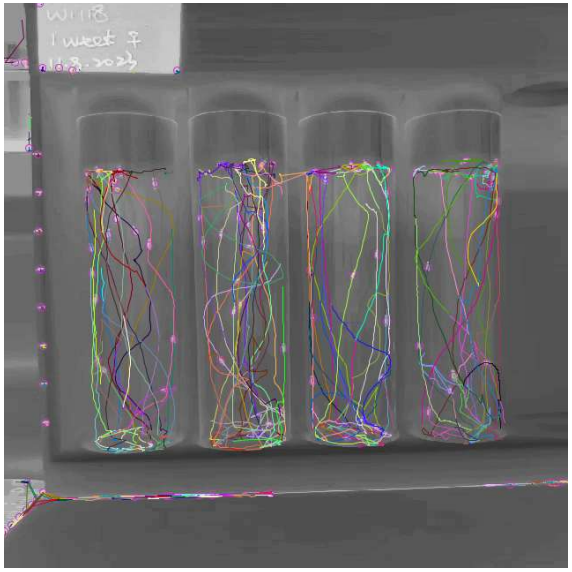


LAP tracker

Exercise 2

Open Flies_tracking_cropped.tif
Convert the image to grayscale 8bit (Image > Type > 8-bit)
Invert the image to get bright flies on a dark background (Edit > Invert)
Start Trackmate (Plugins>Tracking>Trackmate)
Use a rectangle to reduce the XYZ area being used for tracking (remove unnecessary frame from the movie)
Use a blob detector to detect the flies (tune cell size and quality to optimize the detection)
Use the simple LAP tracker to track the flies





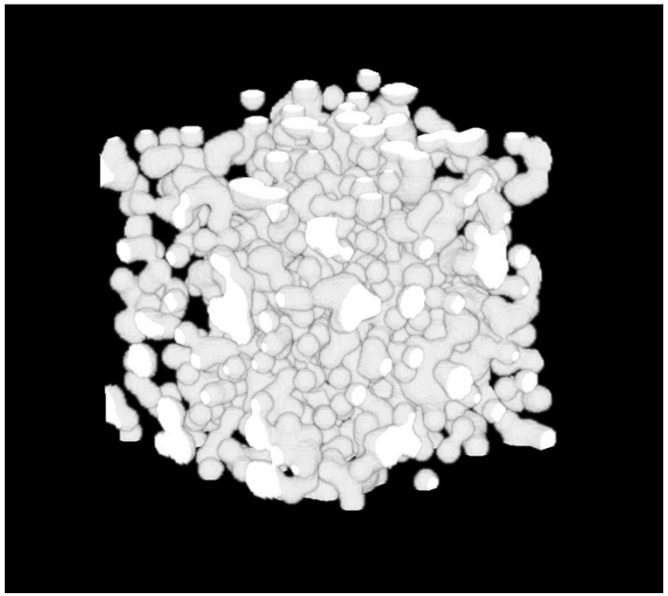
Questions ?

Napari – Basics

1. Open File > Open Sample > Napari > Cells (3D+2ch)



Napari – Basics



- File > Open Samples > napari builtins > Binary Blobs 3D
- By changing the colormap, the interpolation and the rendering, find a way to display smooth 3d blobs (like above)
- Open Plugins > Animation wizard and create your own video (Warning, switching from 2D to 3D may create errors)

- Navigation tips in 3D view:
  - Left click -> Rotation
  - Right click / Mouse Scroll -> Zoom
  - Shift + Left -> Translation
  - Shift + Right -> Weird distortion/perspective

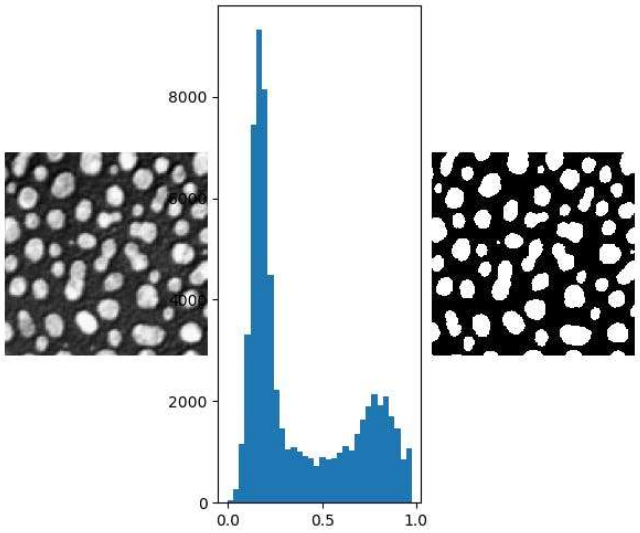
Interpolation : cubic
Depiction : volume
Rendering : attenuated_mip
Attenuation : 0.2

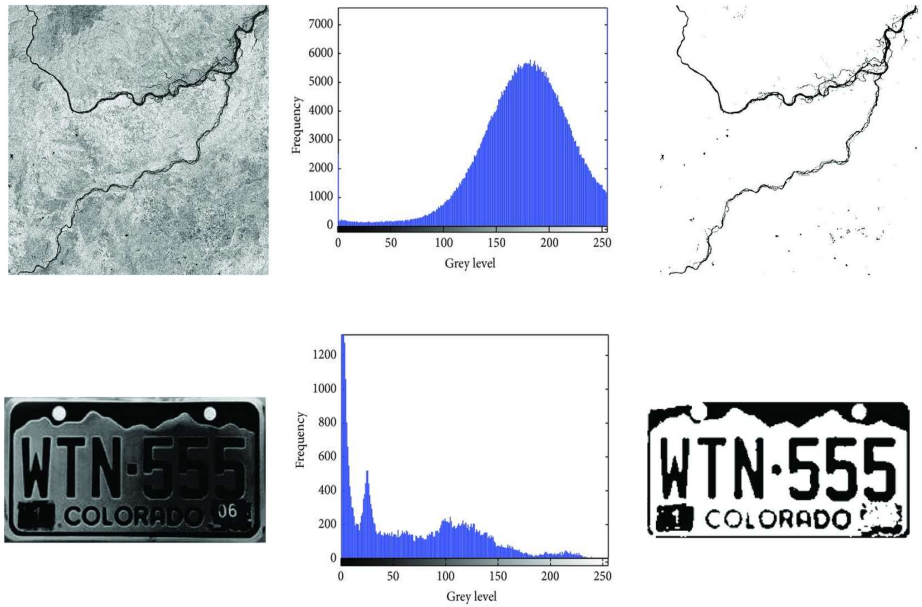
https://napari.org/stable/tutorials/fundamentals/installation.html

Napari step 1 - Preprocessing

- File > Open Samples > Napari > Cell (3D+2Ch)

- Before thresholding an image, it is a common practice to remove the noise from an image to obtain a better result
- Under Tools > Noise removal, apply a Median filter (n-SimpleITK) of 3x3x3 on the nuclei layer
- In which case would you use a Median Filter instead of a Gaussian filter ?
- Binarize the image using the Kittler & Illingworth threshold method.



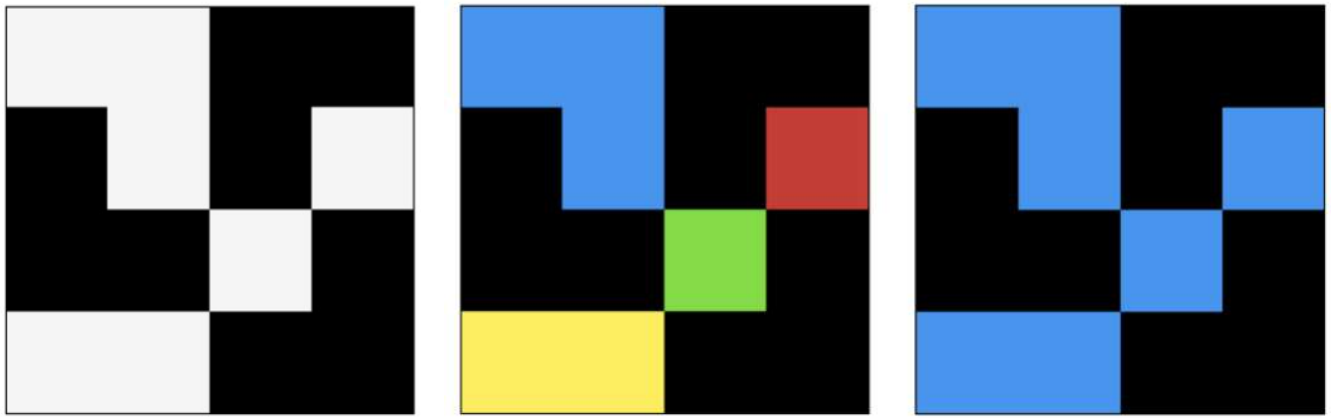


Kittler & Illingworth, Pattern Recognition (1986)

Jin Liu & al, Mathematical Problems in Engineering (2014)

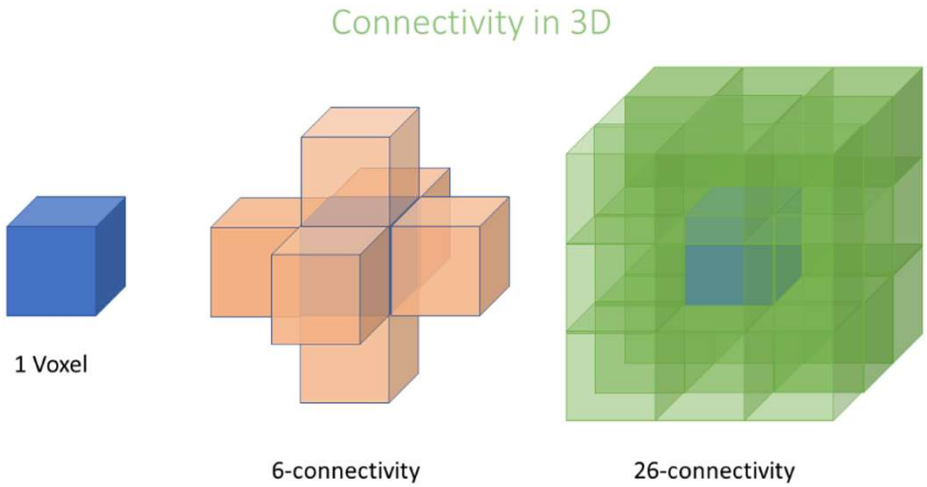
Napari step 2 - Segmentation

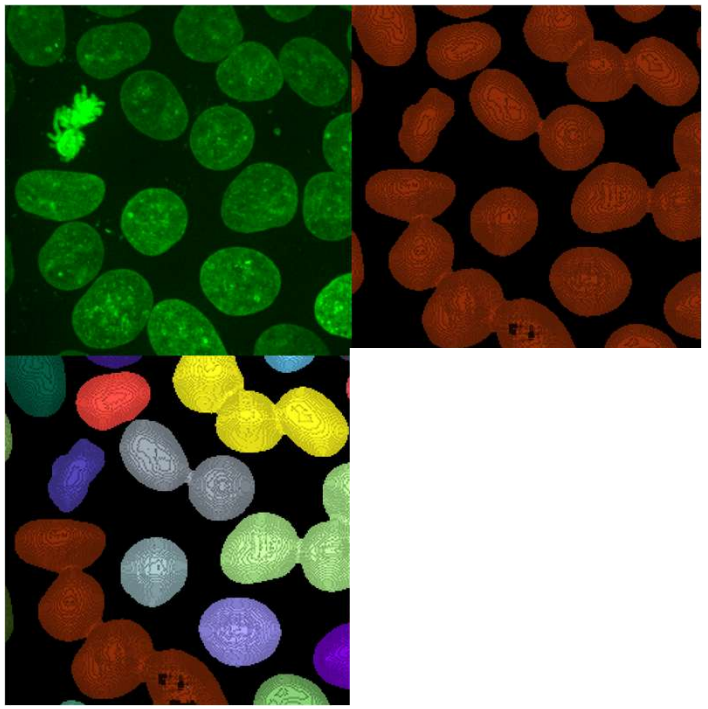
- To improve the segmentation, use the *segmentation post-processing* methods *Binary Fill Holes*
- Create labels. This will give an index and a different colour at each nuclei.



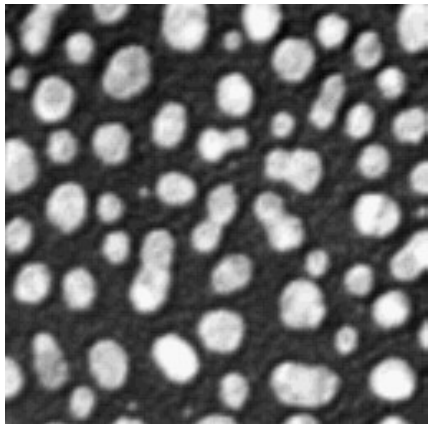
Napari step 2 - Segmentation

- To improve the segmentation, use the *segmentation post-processing* methods *Binary Fill Holes*
- Create labels. This will give an index and a different colour at each nuclei.

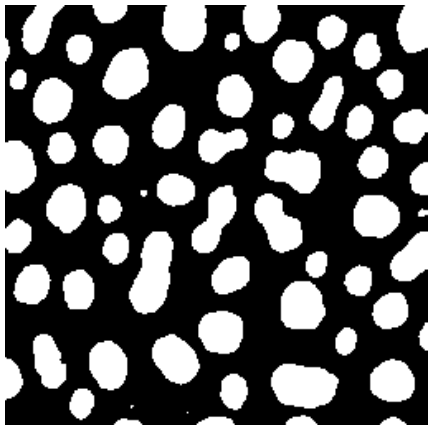




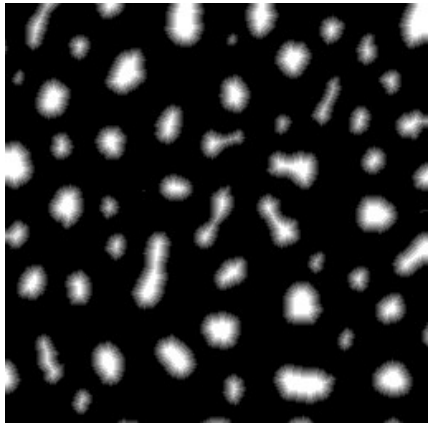
Napari step 2 – Segmentation (Watershed)



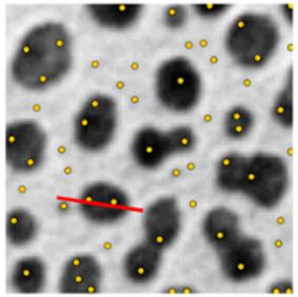
Blobs

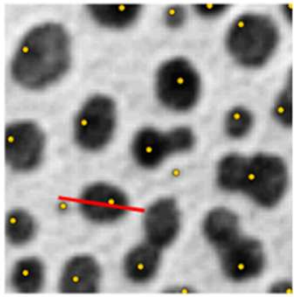


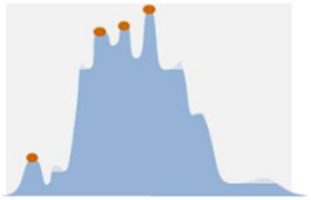
Binary



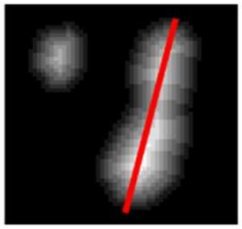
Distance Map

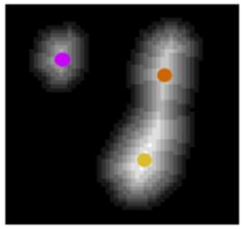


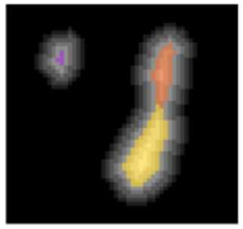


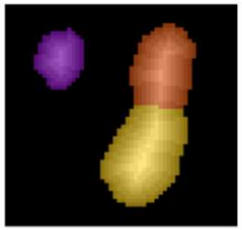


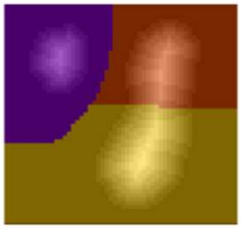




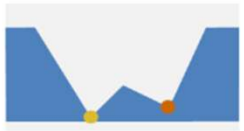


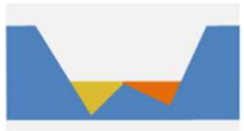






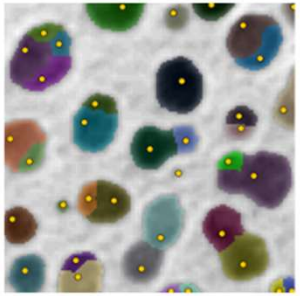


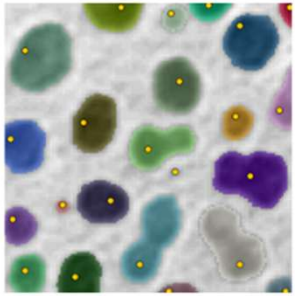








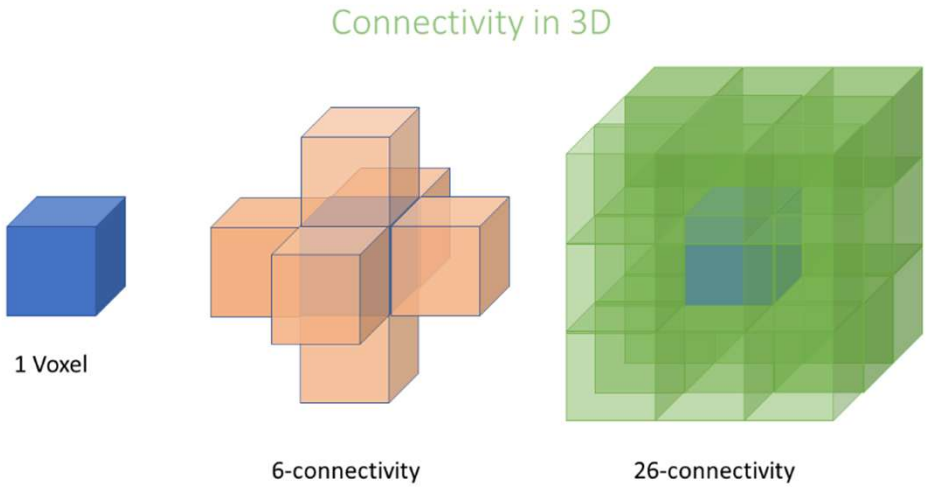


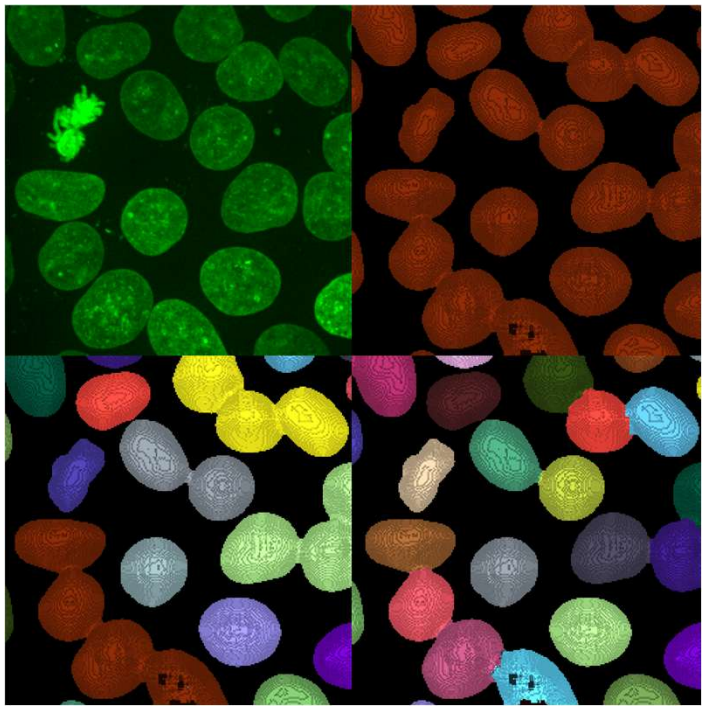


Prominence

Napari step 2 - Segmentation

- To improve the segmentation, use the *segmentation post-processing* methods *Binary Fill Holes*
- Create labels. This will give an index and a different colour at each nuclei.
- Create labels using *Touching object labeling*. This will give an index and a different colour at each nuclei.





Napari step 3 - Measurement

- Reduce the median filter to a size of 3x3x3, and *Binarize* the image using the *Kittler & Illingworth* threshold method.
- To improve the segmentation, use the *segmentation post-processing* methods *Binary Fill Holes*
- Create labels using *Touching object labeling*. This will give an index and a different colour at each nuclei.
- Create a *Measurement tables* using *Measurements (n-SimpleITK)*
  - Pick the raw intensity image (*nuclei*) and the latest label image (*Result of touching_objects*)
  - Tick size, intensity, shape
- Create a new measurement map layer by double-clicking the column header of the table
  - For example, double-click on *mean* column to generate an intensity map of each nuclei
  - You can also create some Measurement maps under the *Measurement maps* menu (fx Pixel Count map (n-SimpleITK))

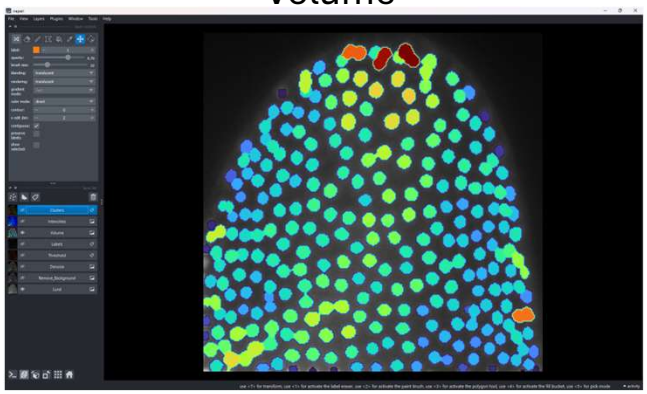
- Classify the nuclei by using Tools > Measurement > Plot Measurements
  - Select the *Result of touching_objects* label layer, plot the *mean* versus *number_of_pixels* data
  - Use *left-click* to classify nuclei using freehand
  - Use *right-click* to classify nuclei using square
  - Hold *Shift* while classifying to assign the nuclei to a new class

- Start the *script editor* from the *Plugins* menu
  - Tick and untick record to generate the script of your workflow.
  - Save the script generated to a new python file for later use

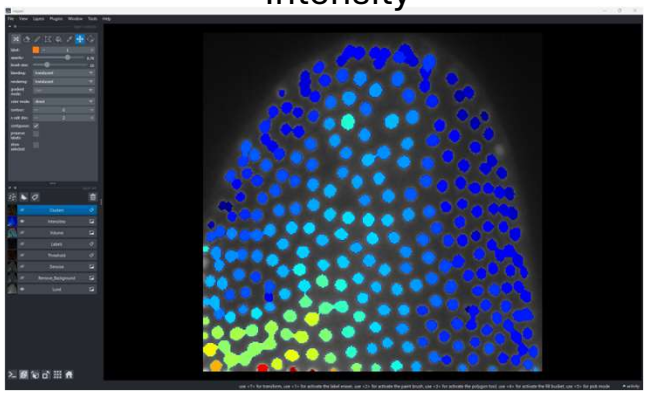
Napari Extras

- Try to replicate a similar workflow using *File* > *Open Samples* > *clEsperanto* > *Lund*
  - Hint : An first additional step of *Background removal* using the *White-top-hat* algorithm is necessary (with default parameters)

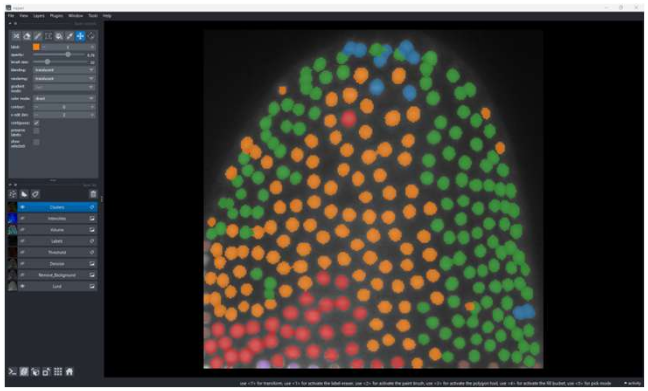
Volume



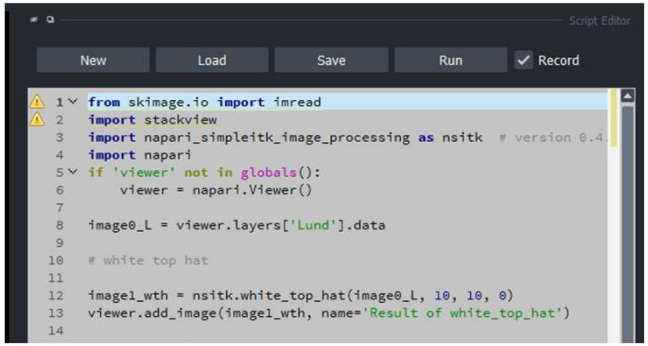
Intensity



Clusters

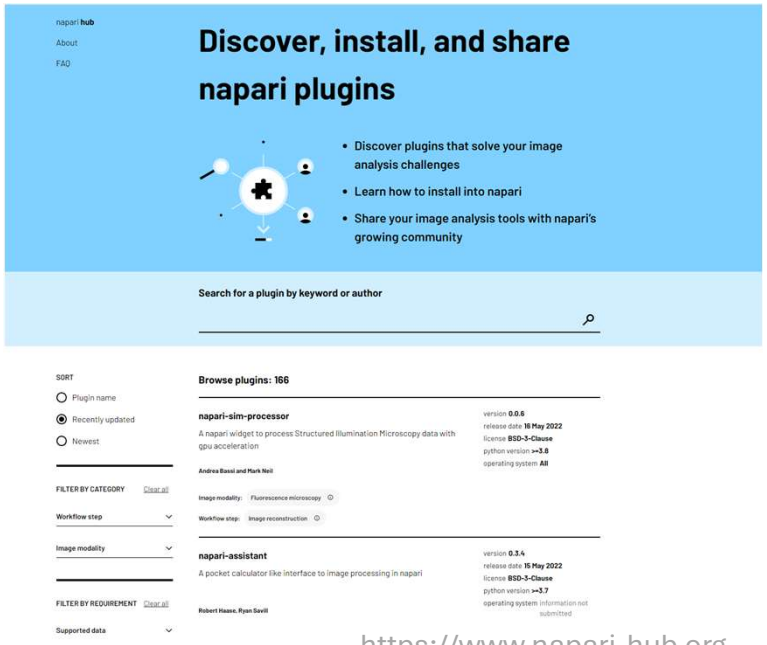


- Once finished, start the *script editor* from the *Plugins* menu
  - Tick and untick *record* to generate the script of your workflow.
  - Save the script generated to a new python file for later use



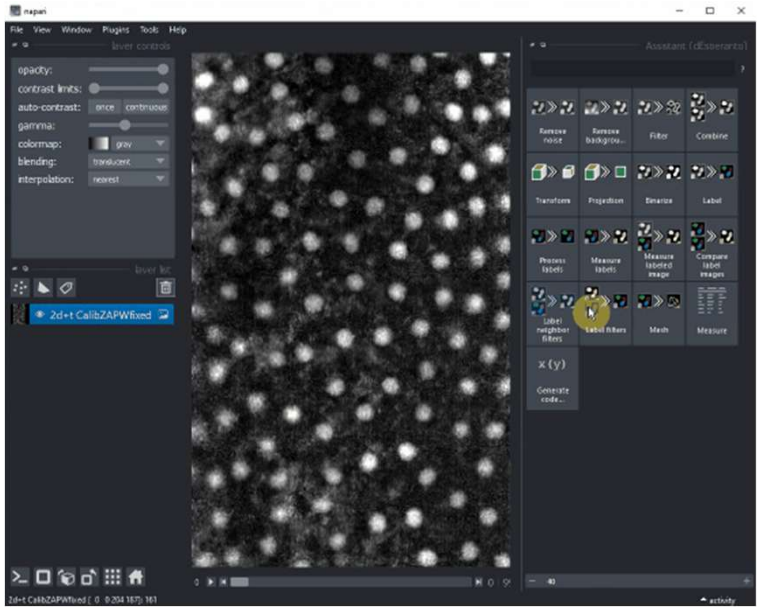
Napari Extras

- Powerful, fast
- Good rendering
- Not intuitive for non-python coders, but very versatile



https://www.napari-hub.org

pyclesperanto



Clusters-plotter

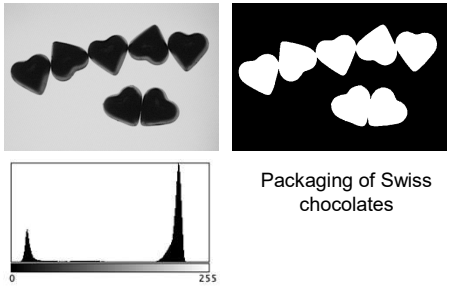


Questions ?

Rule-based

Intensity-based
Ad hoc routines
Aggregation pixels

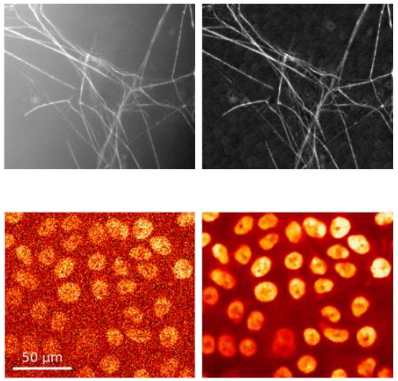
Preprocessing
Handcrafted filters



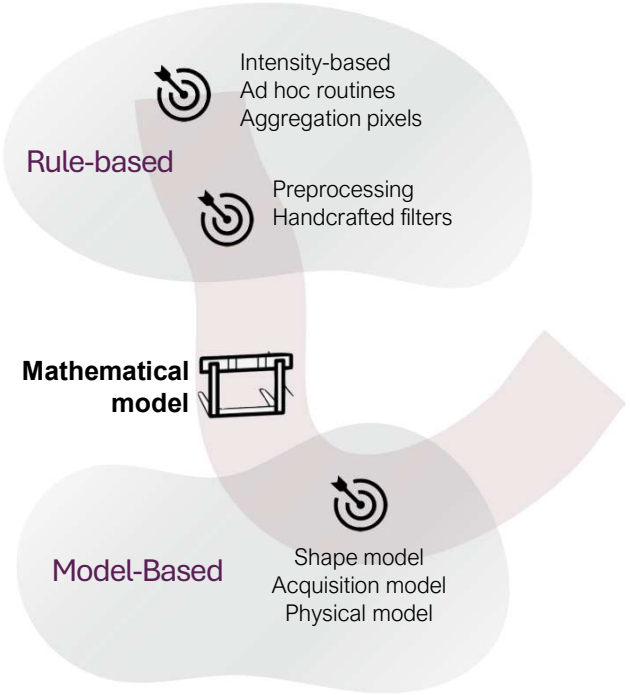
Packaging of Swiss chocolates

Preprocessing

- Frequency filtering
- Denoising
- Projection over time
- Flatten background

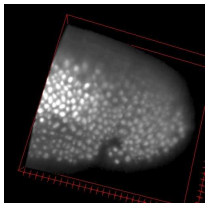


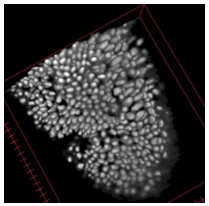
Daniel Sage, Estibaliz Gomez de Mariscal , Neubias 2023



Physic-driven

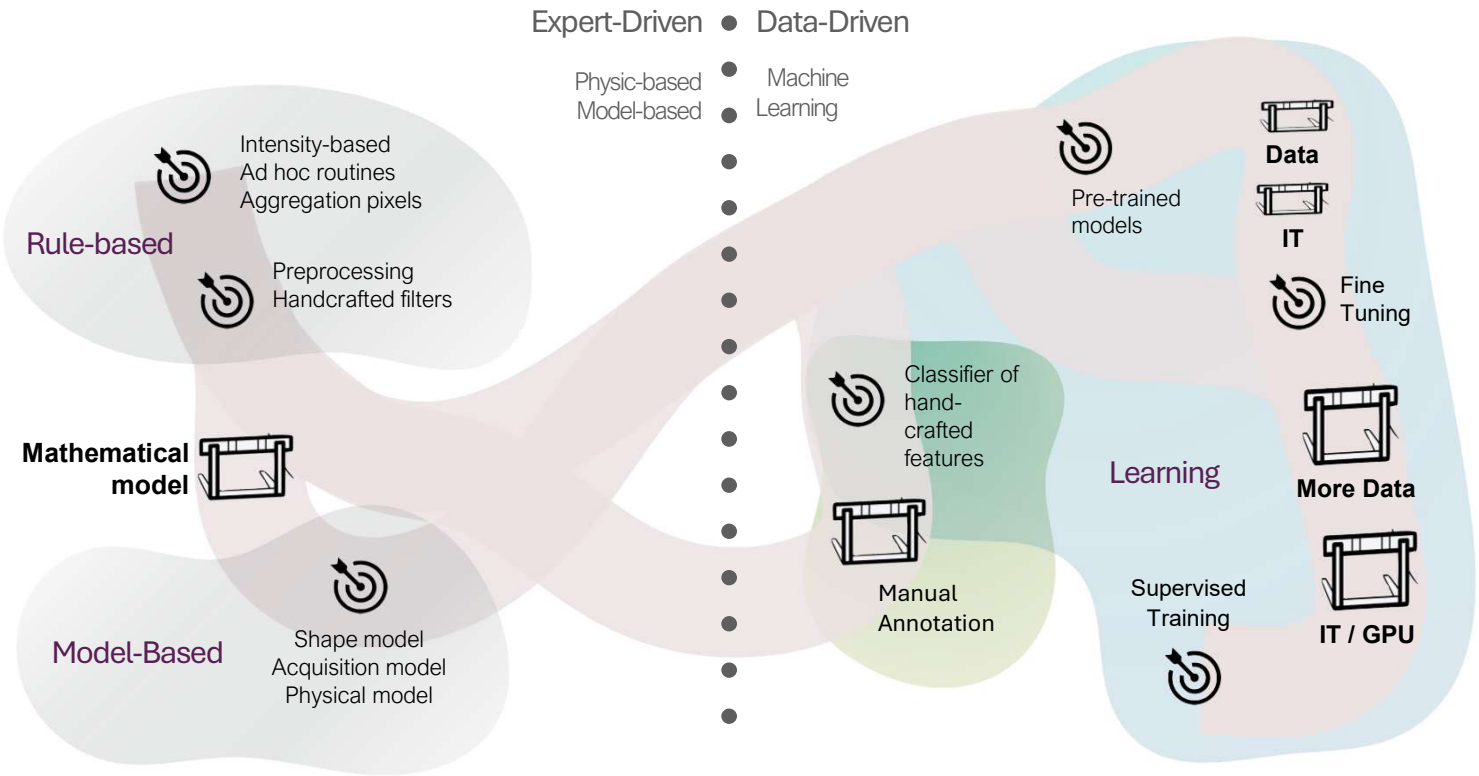
- Deconvolution
- Correlative imaging
- Image reconstruction
- Super-resolution mic.



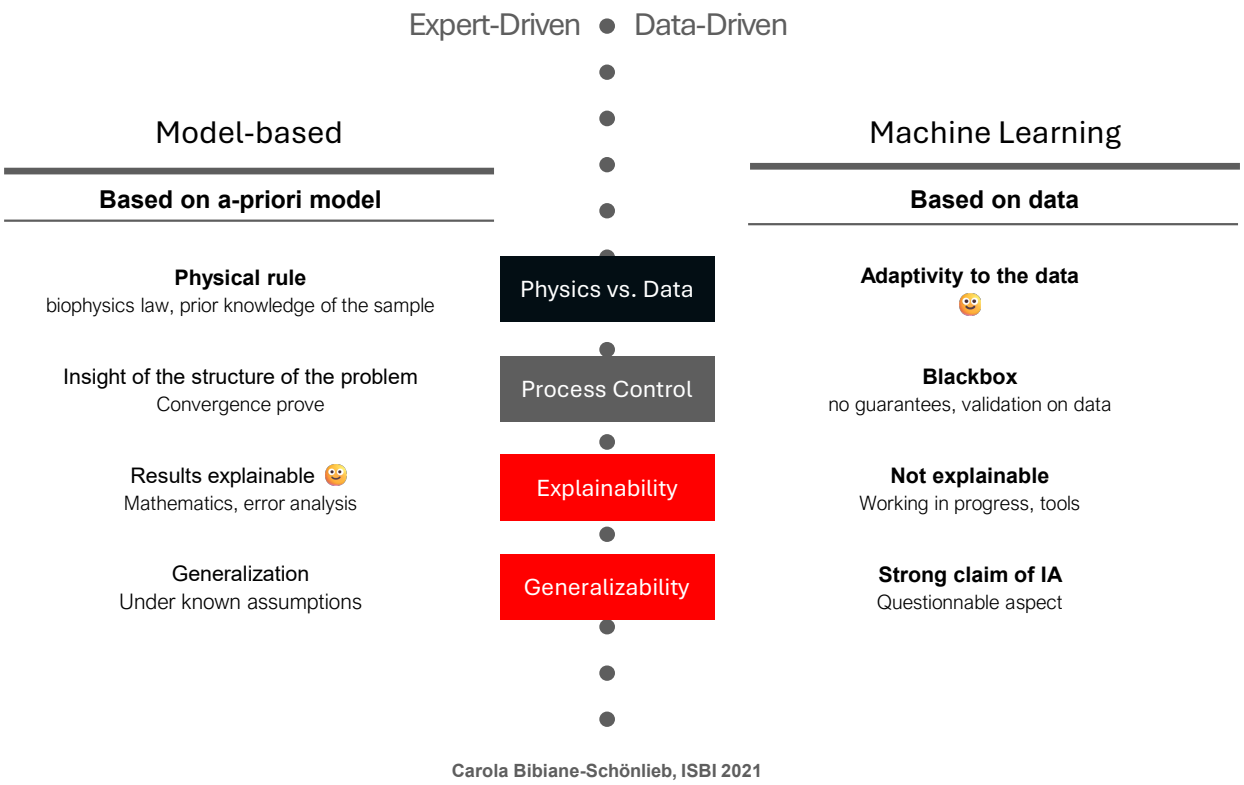


Sage et al. DeconvolutionLab2 Methods 2017.

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